QRISK_meta_analysis

install.packages("readxl") install.packages("meta") install.packages("metafor") install.packages("dplyr") install.packages("ggplot2")

IMPORT DATA FROM EXCEL FILE

```r
# Inport the data from the xcel fille
# Load package
library(readxl)

# Google Drive file ID (uploaded .xlsx, not a native Sheet)
sheet_id <- "1yJQ9Pd9yODPqh9ewfctuPeNe1Gdx7mbg"
url <- paste0("https://drive.google.com/uc?export=download&id=", sheet_id)

tmp <- tempfile(fileext = ".xlsx")
download.file(url, tmp, mode = "wb")     # mode = "wb" is essential
df <- read_excel(tmp)

# View first few rows
head(df)
```

```
## # A tibble: 6 x 41
##   Number Index Author   Title Sample_size Study_Year Disease_Group QRISK_Version
##    <dbl> <chr> <chr>    <chr>       <dbl>      <dbl> <chr>         <chr>
## 1      1 1     Enguita~ 01 E~       20793       2025 "Diabetes (T~ QRISK2
## 2      3 3     Azoulay~ 03 P~         143       2025 "Systemic Lu~ QRISK3
## 3      4 4     Krishna~ 04 C~        5371       2024 "Diabetes (T~ QRISK3
## 4      5 5a    Llorca ~ 05 P~         708       2012 "\r\n Rheuma~ QRISK3
## 5      5 5b    Llorca ~ 05 P~         692       2012 "Ankylosing ~ QRISK3
## 6      5 5c    Llorca ~ 05 P~         680       2012 "Psoriatic A~ QRISK3
## # i 33 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## #   `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## #   Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## #   Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## #   QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## #   Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## #   Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...
```

```r
# Check structure
str(df)
```

```
## tibble [34 x 41] (S3: tbl_df/tbl/data.frame)
##  $ Number                : num [1:34] 1 3 4 5 5 5 6 6 6 7 ...
```

```
##  $ Index                : chr [1:34] "1" "3" "4" "5a" ...
##  $ Author               : chr [1:34] "Enguita-German et al. (T2DM-2025)" "Azoulay et al. (SLE-2025)
##  $ Title                : chr [1:34] "01 External validation of cardiovascular risk scores?in patie
##  $ Sample_size          : num [1:34] 20793 143 5371 708 692 ...
##  $ Study_Year           : num [1:34] 2025 2025 2024 2012 2012 ...
##  $ Disease_Group        : chr [1:34] "Diabetes (T2DM)" "Systemic Lupus Erythematosus (SLE)" "Diabet
##  $ QRISK_Version        : chr [1:34] "QRISK2" "QRISK3" "QRISK3" "QRISK3" ...
##  $ Outcome              : chr [1:34] "CV death, non-fatal MI, non-fatal stroke" "The primary outcom
##  $ AUC                  : chr [1:34] "0.663 (0.646 - 0.681)" "0.76 (0.57-0.93)" "0.788 (0.713-
0.863)" "0.60660000000000003" ...
##  $ Calibration          : chr [1:34] "CITL men and women -1.292 -1.455 (type 1 recalibration)    \r
Underestimated.\r\nPsA - Underestimated\r\nAS - Underestimated\r\nHR 1.04 (1.03-1.05)" ...
##  $ Follow-Up            : chr [1:34] "5 years" "7 years (IQR 3-9 years)" "1 - 12 years\r\nLongitudi
##  $ Country              : chr [1:34] "Spain" "France" "India" "Spain" ...
##  $ Participants         : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ Predictors2          : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ Outcome3             : chr [1:34] "HIGH" "HIGH" "HIGH" "LOW" ...
##  $ Analysis             : chr [1:34] "LOW" "HIGH" "HIGH" "UNCLEAR" ...
##  $ Overall              : chr [1:34] "HIGH" "HIGH" "HIGH" "UNCLEAR" ...
##  $ Metanalysis          : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
##  $ Confidence_interval  : chr [1:34] "Pooled" "Reported" "Approximated" "Not _reported" ...
##  $ Dscrimination        : chr [1:34] "Harrell's C-index" "AUC" "AUC" "Heller's K" ...
##  $ QRISK_2              : num [1:34] 1 0 0 0 0 0 0 0 0 0 ...
##  $ QRISK_3              : num [1:34] 0 1 1 1 1 1 1 1 1 1 ...
##  $ Outcome_type         : chr [1:34] "Appropriate" "Appropriate" "Somewhat_Appropriate" "Appropriat
##  $ Outcome_Description  : chr [1:34] "CVD" "CVD" "CVD( not including peripheral arte-\r\nrial disea
##  $ Applicability        : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
##  $ Include              : num [1:34] 1 1 1 1 1 1 1 1 1 1 ...
##  $ PA                   : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ PR                   : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ OT                   : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ OA                   : chr [1:34] "LOW" "LOW" "LOW" "LOW" ...
##  $ Study_2              : chr [1:34] "01 Enguita-German" "03 Azoulay 2025 (Lupus)" "04 Aswath Krish
##  $ Single_vs_multi-centre: chr [1:34] "Multi-centre" "Single-centre" "Single-centre" "Multi-centre"
##  $ Events               : num [1:34] 991 12 50 67 50 57 607 70 485 34 ...
##  $ Mean_age             : chr [1:34] "63.5" "51" "Not reported for whole cohort (low-risk 38.2, hig
##  $ SD_age               : chr [1:34] "10.8" "Not provided" "Not reported for whole cohort (low-risk
##  $ Male                 : chr [1:34] "12225" "13" "486" "Not reported by disease subgroup (whole co
##  $ Female               : num [1:34] 8568 130 4885 NA NA ...
##  $ Sensitivity_clean    : chr [1:34] "Not provided" "83%" "0.78" "Not provided" ...
##  $ Specificity_clean    : chr [1:34] "Not provided" "71%" "0.68999999999999995" "Not provided" ...
##  $ Calibration_shot     : chr [1:34] "OVER-estimation of risk in original model; good calibration o
```

```
# Check column names
names(df)
```

```
##  [1] "Number"               "Index"                "Author"
##  [4] "Title"                "Sample_size"          "Study_Year"
##  [7] "Disease_Group"        "QRISK_Version"        "Outcome"
## [10] "AUC"                  "Calibration"          "Follow-Up"
## [13] "Country"              "Participants"         "Predictors2"
## [16] "Outcome3"             "Analysis"             "Overall"
## [19] "Metanalysis"          "Confidence_interval"  "Dscrimination"
```

```
## [22] "QRISK_2"               "QRISK_3"               "Outcome_type"
## [25] "Outcome_Description"   "Applicability"         "Include"
## [28] "PA"                    "PR"                    "OT"
## [31] "OA"                    "Study_2"               "Single_vs_multi-centre"
## [34] "Events"                "Mean_age"              "SD_age"
## [37] "Male"                  "Female"                "Sensitivity_clean"
## [40] "Specificity_clean"     "Calibration_shot"
```

CLEAN THE IMPORTED DATA BY REMOVING ANY EMPTY COLUMNS THE RECHECK DATA

```r
# Remove columns that are entirely NA
df <- df[, colSums(!is.na(df)) > 0]

# Check dimensions
dim(df)
```

```
## [1] 34 41
```

```r
# Check column names
names(df)
```

```
##  [1] "Number"                "Index"                 "Author"
##  [4] "Title"                 "Sample_size"           "Study_Year"
##  [7] "Disease_Group"         "QRISK_Version"         "Outcome"
## [10] "AUC"                   "Calibration"           "Follow-Up"
## [13] "Country"               "Participants"          "Predictors2"
## [16] "Outcome3"              "Analysis"              "Overall"
## [19] "Metanalysis"           "Confidence_interval"   "Dscrimination"
## [22] "QRISK_2"               "QRISK_3"               "Outcome_type"
## [25] "Outcome_Description"   "Applicability"         "Include"
## [28] "PA"                    "PR"                    "OT"
## [31] "OA"                    "Study_2"               "Single_vs_multi-centre"
## [34] "Events"                "Mean_age"              "SD_age"
## [37] "Male"                  "Female"                "Sensitivity_clean"
## [40] "Specificity_clean"     "Calibration_shot"
```

REMOVE COMPLETELY EMPTY ROWS

```r
# Remove rows where all values are NA
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
df <- df %>%
  filter(!if_all(everything(), is.na))
```

CHECK THE FORMAT OF INPUTED CONFIDENCE INTERVAL IN ORDER TO DEVICE PLAN TO PARSE IT

```r
# check CI format
df$AUC[!is.na(df$AUC)][1:20]
```

```
##  [1] "0.663 (0.646 - 0.681)"           "0.76 (0.57-0.93)"
##  [3] "0.788 (0.713-0.863)"             "0.60660000000000003"
##  [5] "0.59189999999999998"             "0.57999999999999996"
##  [7] "0.74 (0.72-0.76)"                "\r\n0.70 (0.64-0.76)"
##  [9] "0.716 (0.69-0.74)"               "0.703(0.603-0.804)"
## [11] "\r\n\r\n\r\n0.752 (0.734-0.777)" "\r\n\r\n0.794 (0.764-0.812)"
## [13] "0.764 (0.741-0.791)"             "\r\n\r\n0.815 (0.789-0.835)"
## [15] "0.754 (0.64-0.87)"               "0.878 (0.841-0.915)"
## [17] "0.765 (0.711-0.820)"             "0.664 (0.660 - 0.667)"
## [19] "0.664 (0.660 - 0.668)"           "0.697 (0.659-0.736)"
```

SPLIT CONFIDENCE INTERVAL

CLEAN AND EXTRACT NUMERICAL VALUES FROM AUC COLUMN

```r
# extract CI into lower and upper column
library(stringr) # Provides fuction for manipulating text
library(dplyr) # provides function for manipulating data

df <- df %>% # use the pipe operator to pass the result of manipulation and passes it to
↪ the right
  mutate( # create new columns while retaining exisiting ones
    AUC_clean = str_squish(AUC), #remove unnecessary spaces from AUC and store it in new
     ↪ variable

    Estimate = str_extract(AUC_clean, "^\\d*\\.?\\d+") |> as.numeric(), # creates a
     ↪ variabl, Estimate and store the first number seen in the AUC_clean is it as a
     ↪ number

    LowerCI = str_match( # Using regular expression extract the lower confidence interval
     ↪ and store it in LowerCI column
      AUC_clean,
      "\\(\\s*(\\d*\\.?\\d+)\\s*[--]\\s*(\\d*\\.?\\d+)\\s*\\)"
    )[,2] |> as.numeric(),

    UpperCI = str_match( # Using regular expression extract the Upper confidence interval
     ↪ and store it in UpperCI column
      AUC_clean,
      "\\(\\s*(\\d*\\.?\\d+)\\s*[--]\\s*(\\d*\\.?\\d+)\\s*\\)"
    )[,3] |> as.numeric()
  )
```

```r
# view all the new extracted data
df %>%
  select(AUC, Estimate, LowerCI, UpperCI) %>%
  head(30)
```

```
## # A tibble: 30 x 4
##    AUC                        Estimate LowerCI UpperCI
##    <chr>                         <dbl>   <dbl>   <dbl>
##  1 "0.663 (0.646 - 0.681)"    0.663   0.646   0.681
##  2 "0.76 (0.57-0.93)"         0.76    0.57    0.93
##  3 "0.788 (0.713-0.863)"      0.788   0.713   0.863
##  4 "0.60660000000000003"      0.607  NA      NA
##  5 "0.59189999999999998"      0.592  NA      NA
##  6 "0.57999999999999996"      0.58   NA      NA
##  7 "0.74 (0.72-0.76)"         0.74    0.72    0.76
##  8 "\r\n0.70 (0.64-0.76)"     0.7     0.64    0.76
##  9 "0.716 (0.69-0.74)"        0.716   0.69    0.74
## 10 "0.703(0.603-0.804)"       0.703   0.603   0.804
## # i 20 more rows
```

HANLEY & MCNEIL (1982) APPROXIMATION

```r
# HANLEY & MCNEIL (1982) APPROXIMATION
library(dplyr)

df_hm <- df # create a working copy

df_hm <- df_hm %>%
  mutate( # create a new column
    n1 = as.numeric(Events), # store the information in the Event column in n1 as number
    n0 = Sample_size - n1, # compute the number of none events by subtracting event from
     ↪ sample size
    Q1 = Estimate / (2 - Estimate), # compute for Q1
    Q2 = 2 * Estimate^2 / (1 + Estimate), # compute for Q2
    HM_se    = sqrt((Estimate * (1 - Estimate) + # h and M original variance estimator
                     (n1 - 1) * (Q1 - Estimate^2) +
                     (n0 - 1) * (Q2 - Estimate^2)) / (n1 * n0)),
    HM_lower = pmax(0, Estimate - 1.96 * HM_se), # calcualte lower confidence interval
    HM_upper = pmin(1, Estimate + 1.96 * HM_se) # calculate higher confidence interval
  )

# Output the values that where affected by the approximation
hm_table <- df_hm %>% # what would be
  filter(Confidence_interval != "Reported") %>% # isolate row that currently do not have
   ↪ reported confidence interval
  transmute(
    Study    = Index, # indentify the row by the index values
    Events   = n1,
    AUC      = round(Estimate, 3),
    `95% CI` = sprintf("%.3f - %.3f", HM_lower, HM_upper)
  )
```

```r
print(as.data.frame(hm_table), row.names = FALSE)
```

```
##  Study Events   AUC        95% CI
##      1    991 0.663 0.644 - 0.682
##      4     50 0.788 0.713 - 0.863
##     5a     67 0.607 0.532 - 0.681
##     5b     50 0.592 0.506 - 0.678
##     5c     57 0.580 0.500 - 0.660
##      7     34 0.703 0.602 - 0.804
##     14     26 0.754 0.639 - 0.869
##     16    138 0.878 0.839 - 0.917
##    26a    232 0.720 0.681 - 0.759
##    26b    232 0.670 0.630 - 0.710
##     32      9 0.510 0.317 - 0.703
```

CALCULATE STANDARD ERROR

```r
# STANDARD ERROR
df$SE <- (df$UpperCI - df$LowerCI) / 3.92
```

```r
# VERIFY NO VLAUES ARE MISSING
sum(is.na(df$SE))
```

```
## [1] 3
```

```r
summary(df$SE)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.     NA's
## 0.001786 0.011990 0.020408 0.031998 0.040816 0.096939        3
```

SPLIT DATA (df_all, df_primary)

df_all : dataframe containing all included studies

df_primary : dataframe containing only studies included in the primary pool

```r
# Load the package
library(writexl)
```

```r
# create df with all data including QRISK
df_all <- df

nrow(df_all)
```

```
## [1] 34
```

```r
head(df_all)
```

```
## # A tibble: 6 x 46
##   Number Index Author   Title Sample_size Study_Year Disease_Group QRISK_Version
##    <dbl> <chr> <chr>    <chr>       <dbl>      <dbl> <chr>         <chr>
## 1      1 1     Enguita~ 01 E~       20793       2025 "Diabetes (T~ QRISK2
## 2      3 3     Azoulay~ 03 P~         143       2025 "Systemic Lu~ QRISK3
## 3      4 4     Krishna~ 04 C~        5371       2024 "Diabetes (T~ QRISK3
## 4      5 5a    Llorca ~ 05 P~         708       2012 "\r\n Rheuma~ QRISK3
## 5      5 5b    Llorca ~ 05 P~         692       2012 "Ankylosing ~ QRISK3
## 6      5 5c    Llorca ~ 05 P~         680       2012 "Psoriatic A~ QRISK3
## # i 38 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## #   `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## #   Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## #   Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## #   QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## #   Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## #   Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...
```

```r
# Export the dataframe
write_xlsx(df_all, "df_all.xlsx")
```

Extracting the primary pool studies

```r
# create dataframe for primary pooled AUC
# keep only studies marked for inclusion
df_primary <- df[df$Include == 1, ]

# keep only studies marked for meta - analysis
df_primary <- df_primary[df_primary$Metanalysis == 1, ]

# remove hellers k studies
df_primary <- df_primary[df_primary$Dscrimination != "Heller's K", ]

df_primary <- df_primary[df_primary$Outcome_type == "Appropriate", ]

# View first rows
nrow(df_primary)
```

```
## [1] 19
```

```r
str(df_primary$Index)
```

```
##  chr [1:19] "1" "3" "6a" "6b" "6c" "7" "8a" "8b" "8c" "8d" "16" "19b" "23" ...
```

TRANSFORM EACH C-STATISTIC TO LOGIT SCALE FOR DEBRAY'S ADHERENCE

```r
# transform the AUC values from their original probability scale (0-1) to the logit scale
↪  and calculates the standard error (SE) on the logit scale.
library(dplyr)
```

```r
df_primary_logit <- df_primary %>%
  mutate(
    logit_c  = log(Estimate / (1 - Estimate)), # create logit transformed AUC
    logit_lb = log(LowerCI  / (1 - LowerCI)),
    logit_ub = log(UpperCI  / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96)   # derive the standard error of the
     ↪  logit c-statistic
  )
```

```r
# Export the dataframe
write_xlsx(df_primary_logit, "df_primary-logit.xlsx")
```

```r
# TO CHECK
# output unique studies for primary metanalysis
library(dplyr)
library(stringr)

# Eligible cohorts
eligible <- df_primary_logit %>%
  filter(Metanalysis == 1) %>%
  mutate(
    StudyID = str_extract(as.character(Index), "\\d+")
  )

nrow(eligible)
```

```
## [1] 19
```

```r
cat(
  "Number of unique studies:",
  n_distinct(eligible$StudyID),
  "\n"
)
```

```
## Number of unique studies: 14
```

```r
eligible %>%
  select(Index, Title) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 2
##    Index Title
##    <chr> <chr>
##  1 1     01 External validation of cardiovascular risk scores?in patients with ~
##  2 3     03 Performance of SCORE2, QRISK3 and PREVENT equations in systemic lup~
##  3 6a    06 External validation of the accuracy of cardiovascular risk predicti~
##  4 6b    06 External validation of the accuracy of cardiovascular risk predicti~
##  5 6c    06 External validation of the accuracy of cardiovascular risk predicti~
##  6 7     07 Comparison of Cardiovascular Risk Scores Performances in a Contempo~
```

```
##   7 8a    08 The predictive accuracy of cardiovascular disease risk?prediction t~
##   8 8b    08 The predictive accuracy of cardiovascular disease risk?prediction t~
##   9 8c    08 The predictive accuracy of cardiovascular disease risk?prediction t~
##  10 8d    08 The predictive accuracy of cardiovascular disease risk?prediction t~
##  11 16    16 Assessment of QRISK3 as a predictor of cardiovascular disease event~
##  12 19b   19 Cardiovascular risk prediction in type 2 diabetes/ a comparison of ~
##  13 23    23 Electronic Medical Record Risk Modeling of Cardiovascular Outcomes ~
##  14 26b   26 Assessment of cardiovascular risk tools as predictors of cardiovasc~
##  15 29b   29 Cardiovascular Risk Prediction in Ankylosing Spondylitis/ From Trad~
##  16 35    35 Performances of five risk algorithms in predicting cardiovascular e~
##  17 36    36 Performance of cardiovascular disease risk scores in people diagnos~
##  18 39    39 Comparison of cardiovascular risk algorithms in patients with vs wi~
##  19 40    40 Performance of four current risk algorithms?in predicting cardiovas~
```

Count the number of arms contributed by each study

```
df_primary_logit %>%
  mutate(StudyID = str_extract(as.character(Index), "^[0-9]+")) %>% # CONVERT index to
    ↪ character and then pick out characters from the begining until you find a character
    ↪ that is not a number and store that in studyID
  count(StudyID, name = "n_arms") %>% # get the frequency of each number and store in the
    ↪ column n_arms
  arrange(desc(n_arms)) %>% # arrange the result in decending order
  print(n = Inf) # print all rows
```

```
## # A tibble: 14 x 2
##    StudyID n_arms
##    <chr>    <int>
##  1 8            4
##  2 6            3
##  3 1            1
##  4 16           1
##  5 19           1
##  6 23           1
##  7 26           1
##  8 29           1
##  9 3            1
## 10 35           1
## 11 36           1
## 12 39           1
## 13 40           1
## 14 7            1
```

META-ANALYSIS STARTS

```
# LOAD META PACKAGE
library(meta)
```

```
## Loading required package: metabook
```

```
## Loading 'meta' package (version 8.5-0).
## Type 'help(meta)' for a brief overview.
```

```r
# INVERSE VARIANCE RANDOM EFFECT Using non logit estimates
primary_auc <- metagen( # use metagen package to combine AUC estimates and store in
↪  created object
  TE = Estimate, # effect estimate gotten from Estimate column
  seTE = SE, # get the treatment effect standard error from the SE column
  studlab = Author, # use the Author column to identify studies
  data = df_primary, # dataset that contains all the data
  sm = "AUC", # specify the type of summary measure bein meta-analyzed as AUC
  method.tau = "REML", # how the between study variance is estimated
  method.random.ci = "HK" # method for deriving the confidence interval
)
summary(primary_auc)
```

```
##                                      AUC           95% CI %W(common) %W(random)
## Enguita-German et al. (T2DM-2025) 0.6630 [0.6455; 0.6805]        2.6        6.1
## Azoulay et al. (SLE-2025)         0.7600 [0.5800; 0.9400]        0.0        1.9
## Hughes et al. (Ps-2025)           0.7400 [0.7200; 0.7600]        2.0        6.0
## Hughes et al. (PsA-2025)          0.7000 [0.6400; 0.7600]        0.2        5.0
## Hughes et al. (RA-2025)           0.7160 [0.6910; 0.7410]        1.3        5.9
## Rosário et al. (SLE-2025)         0.7030 [0.6025; 0.8035]        0.1        3.7
## Hughes et al. (RA-2023)           0.7520 [0.7305; 0.7735]        1.8        6.0
## Hughes et al. (AS-2023)           0.7940 [0.7700; 0.8180]        1.4        6.0
## Hughes et al. (PsA-2023)          0.7640 [0.7390; 0.7890]        1.3        5.9
## Hughes et al. (Ps-2023)           0.8150 [0.7920; 0.8380]        1.5        6.0
## Mu et al. (T2DM-2022)             0.8780 [0.8410; 0.9150]        0.6        5.7
## Dziopa K et al. (T2DM-Q3-2022)    0.6640 [0.6600; 0.6680]       50.6        6.2
## Hong et al. (T2DM-2021)           0.7200 [0.7000; 0.7400]        2.0        6.0
## Sivakumaran et al. (SLE-Q3-2021)  0.6700 [0.6300; 0.7100]        0.5        5.6
## Navarini et al. (AS-Q3-2020)      0.6600 [0.4800; 0.8400]        0.0        1.9
## Navarini et al. (PsA-2018)        0.8660 [0.7943; 0.9377]        0.2        4.6
## Read et al. (T2DM-2018)           0.6740 [0.6690; 0.6790]       32.4        6.2
## Alemao et al.(RA-2017)            0.7440 [0.7140; 0.7740]        0.9        5.8
## Arts EEA et al. (RA-2013)         0.7900 [0.7500; 0.8300]        0.5        5.6
##
## Number of studies: k = 19
##
##                              AUC           95% CI    z|t  p-value
## Common effect model       0.6806 [0.6778; 0.6835] 468.78        0
## Random effects model (HK) 0.7420 [0.7104; 0.7736]  49.33 < 0.0001
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.0038 [0.0019; 0.0086]; tau = 0.0617 [0.0432; 0.0930]
##  I^2 = 97.1% [96.3%; 97.7%]; H = 5.86 [5.20; 6.60]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  617.87   18 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
```

```
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
```

```r
# POOLED ESTIMATE ON LOGIT SCALE
library(meta)

primary_logit <- metagen(
  TE               = logit_c, # logit effect estimate gotten from logit_c column
  seTE             = se_logit, # get the logit treatment effect standard error from the
  ↪  Se_logit column
  studlab          = Author, # use the Author column to identify studies
  data             = df_primary_logit, # dataset that contains all the logit transformed
  ↪  data
  method.tau       = "REML",# how the between study variance is estimated
  method.random.ci = "HK", # method for deriving the confidence interval
  prediction       = TRUE # report 95% prediction interval
)
summary(primary_logit)
```

```
##                                                   95% CI %W(common)
## Enguita-German et al. (T2DM-2025) 0.6767 [ 0.5983; 0.7551]        2.8
## Azoulay et al. (SLE-2025)         1.1527 [ 0.0003; 2.3051]        0.0
## Hughes et al. (Ps-2025)           1.0460 [ 0.9419; 1.1501]        1.6
## Hughes et al. (PsA-2025)          0.8473 [ 0.5586; 1.1360]        0.2
## Hughes et al. (RA-2025)           0.9247 [ 0.8018; 1.0476]        1.1
## Rosário et al. (SLE-2025)         0.8616 [ 0.3649; 1.3584]        0.1
## Hughes et al. (RA-2023)           1.1093 [ 0.9927; 1.2259]        1.3
## Hughes et al. (AS-2023)           1.3492 [ 1.2050; 1.4934]        0.8
## Hughes et al. (PsA-2023)          1.1747 [ 1.0348; 1.3146]        0.9
## Hughes et al. (Ps-2023)           1.4828 [ 1.3315; 1.6341]        0.8
## Mu et al. (T2DM-2022)             1.9736 [ 1.6183; 2.3289]        0.1
## Dziopa K et al. (T2DM-Q3-2022)    0.6812 [ 0.6632; 0.6991]       53.7
## Hong et al. (T2DM-2021)           0.9445 [ 0.8451; 1.0438]        1.7
## Sivakumaran et al. (SLE-Q3-2021)  0.7082 [ 0.5266; 0.8898]        0.5
## Navarini et al. (AS-Q3-2020)      0.6633 [-0.2058; 1.5324]        0.0
## Navarini et al. (PsA-2018)        1.8660 [ 1.1856; 2.5465]        0.0
## Read et al. (T2DM-2018)           0.7263 [ 0.7036; 0.7491]       33.3
## Alemao et al.(RA-2017)            1.0669 [ 0.9088; 1.2249]        0.7
## Arts EEA et al. (RA-2013)         1.3249 [ 1.0814; 1.5684]        0.3
##                                   %W(random)
## Enguita-German et al. (T2DM-2025)        6.3
## Azoulay et al. (SLE-2025)                1.5
## Hughes et al. (Ps-2025)                  6.2
## Hughes et al. (PsA-2025)                 5.3
## Hughes et al. (RA-2025)                  6.1
## Rosário et al. (SLE-2025)                3.9
## Hughes et al. (RA-2023)                  6.2
## Hughes et al. (AS-2023)                  6.1
## Hughes et al. (PsA-2023)                 6.1
## Hughes et al. (Ps-2023)                  6.0
## Mu et al. (T2DM-2022)                    4.8
## Dziopa K et al. (T2DM-Q3-2022)           6.4
## Hong et al. (T2DM-2021)                  6.2
```

```
## Sivakumaran et al. (SLE-Q3-2021)         5.9
## Navarini et al. (AS-Q3-2020)             2.2
## Navarini et al. (PsA-2018)               2.9
## Read et al. (T2DM-2018)                  6.4
## Alemao et al.(RA-2017)                   6.0
## Arts EEA et al. (RA-2013)                5.5
##
## Number of studies: k = 19
##
##                                             95% CI    z|t  p-value
## Common effect model       0.7381 [0.7249; 0.7512] 110.13        0
## Random effects model (HK) 1.0631 [0.8899; 1.2363]  12.89 < 0.0001
## Prediction interval              [0.3689; 1.7572]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
```

```r
# invert pooled value and intervals with inverse-logit
plogis(primary_logit$TE.random)      #perform inverse logit transformation on the pooled
↪  estimate
```

```
## [1] 0.7432737
```

```r
plogis(c(primary_logit$lower.random, primary_logit$upper.random))# perform inverse logit
↪  transformation on the confidence interval
```

```
## [1] 0.7088603 0.7749110
```

```r
plogis(c(primary_logit$lower.predict, primary_logit$upper.predict)) # perform inverse
↪  logit transformation in the prediction interval
```

```
## [1] 0.5911950 0.8528587
```

```r
# USING METAMISC LIBRARY AS STATED IN DEBRAY
library(metamisc) # debrey specific package for meta analysis
```

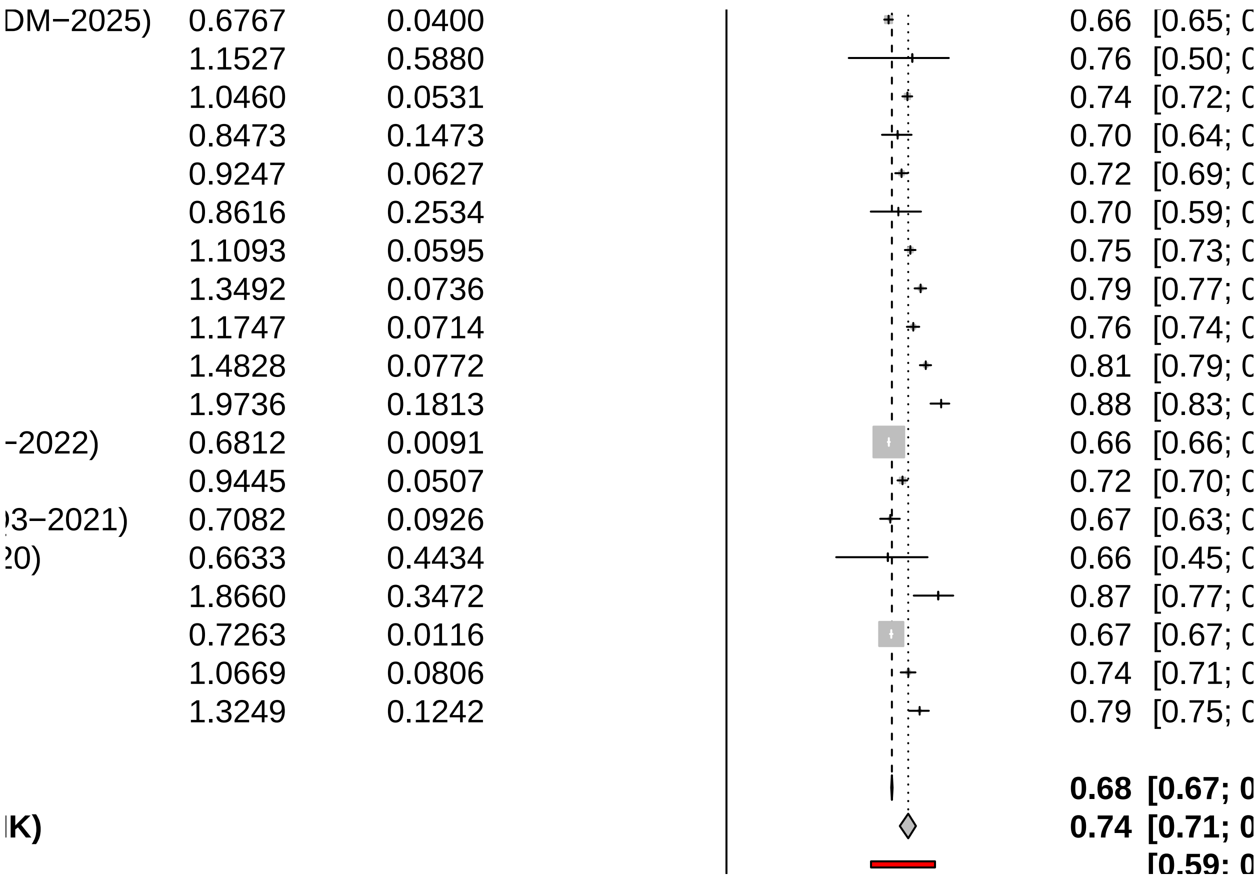
```
##
## Attaching package: 'metamisc'
```

```
## The following object is masked from 'package:meta':
##
##     forest
```

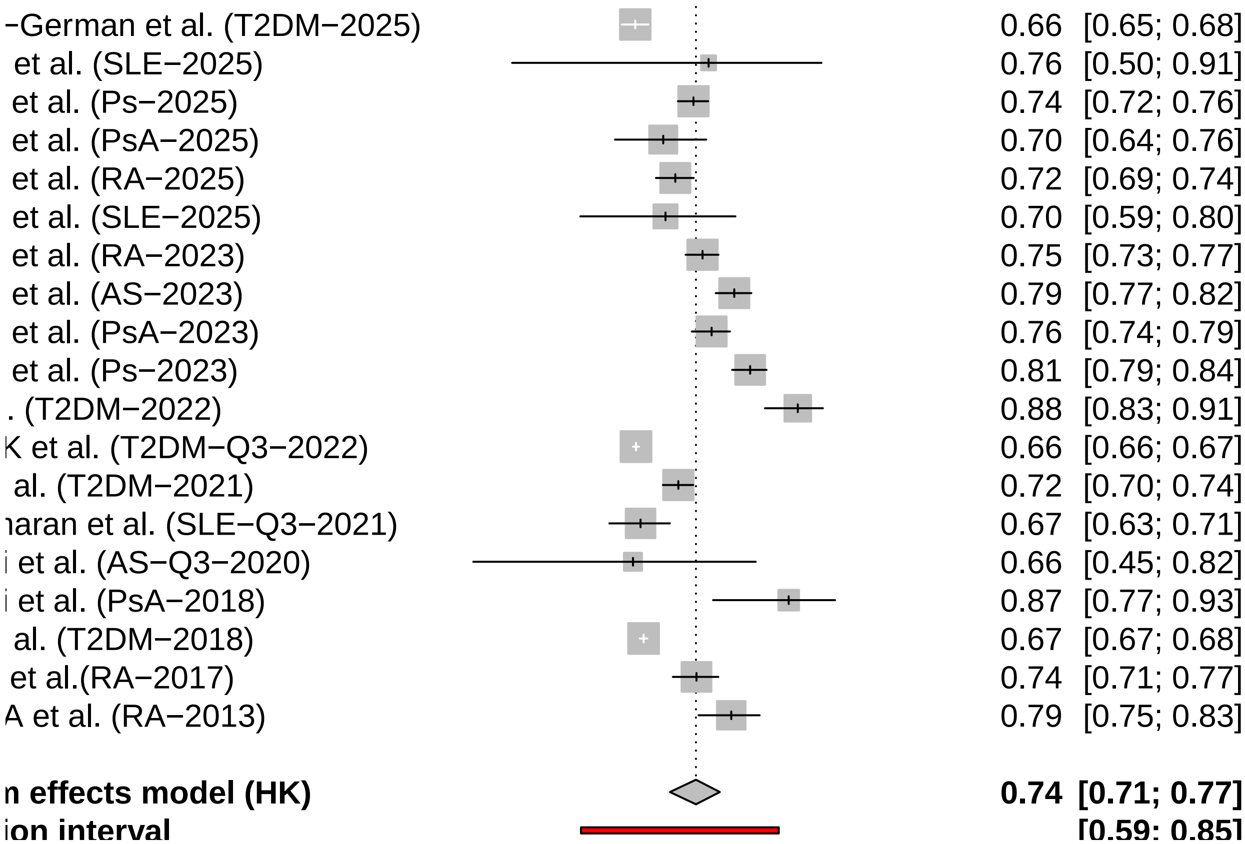
```r
val <- valmeta( # store result of the velmeta function in the val object
  cstat      = df_primary_logit$Estimate,   # RAW c-statistic provided as logit
   ↪ trransformation is done internally
  cstat.cilb = df_primary_logit$LowerCI,  # RAW lower CI
  cstat.ciub = df_primary_logit$UpperCI, # RAW upper CI
  slab       = df_primary_logit$Index, # use index as the study label
  method     = "REML" # how the between study variance is estimated
)
val
```

```
## Summary c-statistic with 95% confidence and (approximate) 95% prediction interval:
##
##  Estimate       CIl       CIu       PIl       PIu
## 0.7432736 0.7088603 0.7749108 0.5902939 0.8533256
##
## Number of studies included:  19
```

```r
# FORREST PLOT VABOSE
library(meta)

primary_logit <- metagen(
  TE               = logit_c, # logit effect estimate gotten from logit_c column
  seTE             = se_logit, # get the logit treatment effect standard error from the
   ↪ Se_logit column
  studlab          = Author, # use the Author column to identify studies
  data             = df_primary_logit, # dataset that contains all the logit transformed
   ↪ data
  sm               = "PLOGIT", # tells meta the scale is logit-transformed proportion
  backtransf       = TRUE, # covert logit values to translatable values
  method.tau       = "REML", # how the between study variance is estimated
  method.random.ci = "HK", # method for deriving the confidence interval
  prediction       = TRUE # report 95% prediction interval
)

meta::forest(primary_logit, xlab = "c-statistic") # use forest function in the meta
 ↪ package
```




```r
# PLOT FORREST PLOT
meta::forest(primary_logit,
             xlab      = "c-statistic",
             xlim      = c(0.4, 1),
             at        = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
             leftcols  = c("studlab"),
             leftlabs  = c("Study"),
             rightcols = c("effect", "ci", "w.random"),
             rightlabs = c("c-statistic", "95% CI", "Weight"),
             common    = FALSE,
             print.I2  = TRUE,
             print.tau2 = TRUE)
```



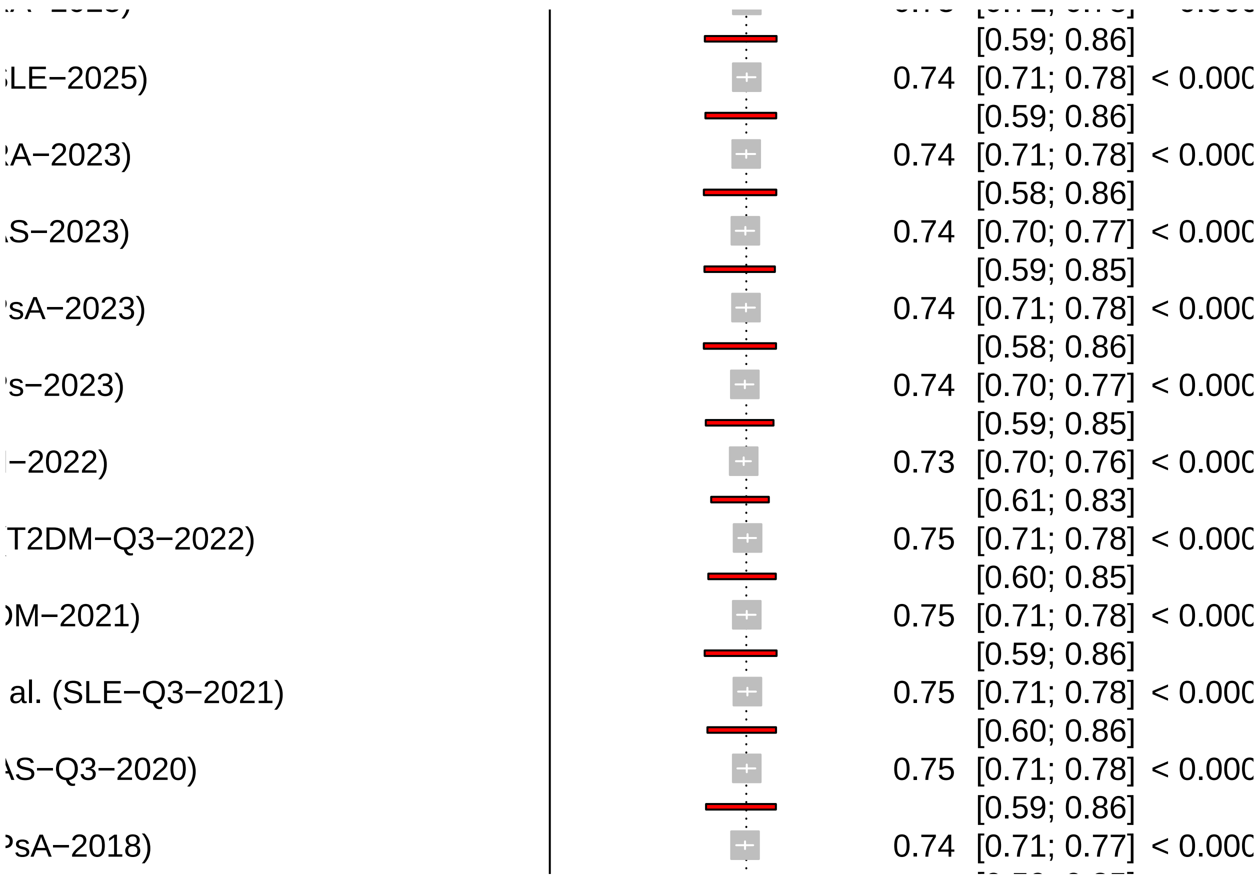
INFLUENCE ANALYSIS

```r
#Influenece analysis to be done on logit transformed estimate
# result will be back transformed for translation
# staring with leave-one-out sensitivity analysis
# If droppping a study shifts the pooled c-statistic by >0.05 the result is not stable
library(meta)
loo <- metainf(primary_logit, pooled = "random") # leave one out
loo
```

```
## Leave-one-out meta-analysis
##
##                                                  proportion           95% CI  p-value
## Omitting Enguita-German et al. (T2DM-2025)           0.7481 [0.7133; 0.7799] < 0.0001
## Omitting Azoulay et al. (SLE-2025)                   0.7431 [0.7072; 0.7760] < 0.0001
## Omitting Hughes et al. (Ps-2025)                     0.7438 [0.7068; 0.7775] < 0.0001
## Omitting Hughes et al. (PsA-2025)                    0.7457 [0.7096; 0.7787] < 0.0001
## Omitting Hughes et al. (RA-2025)                     0.7452 [0.7087; 0.7786] < 0.0001
## Omitting Rosário et al. (SLE-2025)                   0.7450 [0.7090; 0.7780] < 0.0001
## Omitting Hughes et al. (RA-2023)                     0.7430 [0.7060; 0.7767] < 0.0001
## Omitting Hughes et al. (AS-2023)                     0.7398 [0.7035; 0.7730] < 0.0001
## Omitting Hughes et al. (PsA-2023)                    0.7421 [0.7052; 0.7759] < 0.0001
## Omitting Hughes et al. (Ps-2023)                     0.7378 [0.7026; 0.7702] < 0.0001
```

```
## Omitting Mu et al. (T2DM-2022)                0.7334 [0.7041; 0.7609] < 0.0001
## Omitting Dziopa K et al. (T2DM-Q3-2022)       0.7481 [0.7133; 0.7800] < 0.0001
## Omitting Hong et al. (T2DM-2021)              0.7450 [0.7083; 0.7785] < 0.0001
## Omitting Sivakumaran et al. (SLE-Q3-2021)     0.7475 [0.7123; 0.7796] < 0.0001
## Omitting Navarini et al. (AS-Q3-2020)         0.7451 [0.7097; 0.7774] < 0.0001
## Omitting Navarini et al. (PsA-2018)           0.7385 [0.7052; 0.7693] < 0.0001
## Omitting Read et al. (T2DM-2018)              0.7476 [0.7124; 0.7799] < 0.0001
## Omitting Alemao et al.(RA-2017)               0.7435 [0.7066; 0.7772] < 0.0001
## Omitting Arts EEA et al. (RA-2013)            0.7404 [0.7041; 0.7737] < 0.0001
##
## Random effects model (HK)                     0.7433 [0.7089; 0.7749] < 0.0001
##                                             tau^2    tau   I^2           95% PI
## Omitting Enguita-German et al. (T2DM-2025) 0.0987 0.3141 96.1% [0.5995; 0.8549]
## Omitting Azoulay et al. (SLE-2025)         0.1047 0.3236 96.1% [0.5885; 0.8540]
## Omitting Hughes et al. (Ps-2025)           0.1118 0.3343 95.8% [0.5834; 0.8574]
## Omitting Hughes et al. (PsA-2025)          0.1079 0.3284 96.1% [0.5891; 0.8571]
## Omitting Hughes et al. (RA-2025)           0.1103 0.3321 96.0% [0.5865; 0.8578]
## Omitting Rosário et al. (SLE-2025)         0.1074 0.3277 96.1% [0.5886; 0.8564]
## Omitting Hughes et al. (RA-2023)           0.1114 0.3337 95.7% [0.5827; 0.8568]
## Omitting Hughes et al. (AS-2023)           0.1035 0.3217 95.4% [0.5850; 0.8515]
## Omitting Hughes et al. (PsA-2023)          0.1102 0.3319 95.7% [0.5826; 0.8558]
## Omitting Hughes et al. (Ps-2023)           0.0944 0.3073 95.0% [0.5901; 0.8462]
## Omitting Mu et al. (T2DM-2022)             0.0665 0.2579 95.6% [0.6104; 0.8285]
## Omitting Dziopa K et al. (T2DM-Q3-2022)    0.0987 0.3141 95.2% [0.5995; 0.8549]
## Omitting Hong et al. (T2DM-2021)           0.1108 0.3328 96.0% [0.5858; 0.8579]
## Omitting Sivakumaran et al. (SLE-Q3-2021)  0.1018 0.3191 96.1% [0.5962; 0.8558]
## Omitting Navarini et al. (AS-Q3-2020)      0.1046 0.3234 96.1% [0.5910; 0.8553]
## Omitting Navarini et al. (PsA-2018)        0.0944 0.3072 96.0% [0.5912; 0.8465]
## Omitting Read et al. (T2DM-2018)           0.1018 0.3191 96.1% [0.5963; 0.8559]
## Omitting Alemao et al.(RA-2017)            0.1114 0.3338 96.0% [0.5834; 0.8571]
## Omitting Arts EEA et al. (RA-2013)         0.1051 0.3242 95.9% [0.5845; 0.8526]
##
## Random effects model (HK)                  0.1026 0.3203 95.9% [0.5912; 0.8529]
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = {17, 18})
## - Prediction interval based on t-distribution (df = {17, ..., 18})
## - Logit transformation
```

```r
meta::forest(loo, backtransf = TRUE, xlab = "c-statistic (leave-one-out)")
```




```r
library(metafor)
```

```
## Loading required package: Matrix
```

```
## Loading required package: metadat
```

```
## Loading required package: numDeriv
```

```
##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```
##
## Attaching package: 'metafor'
```

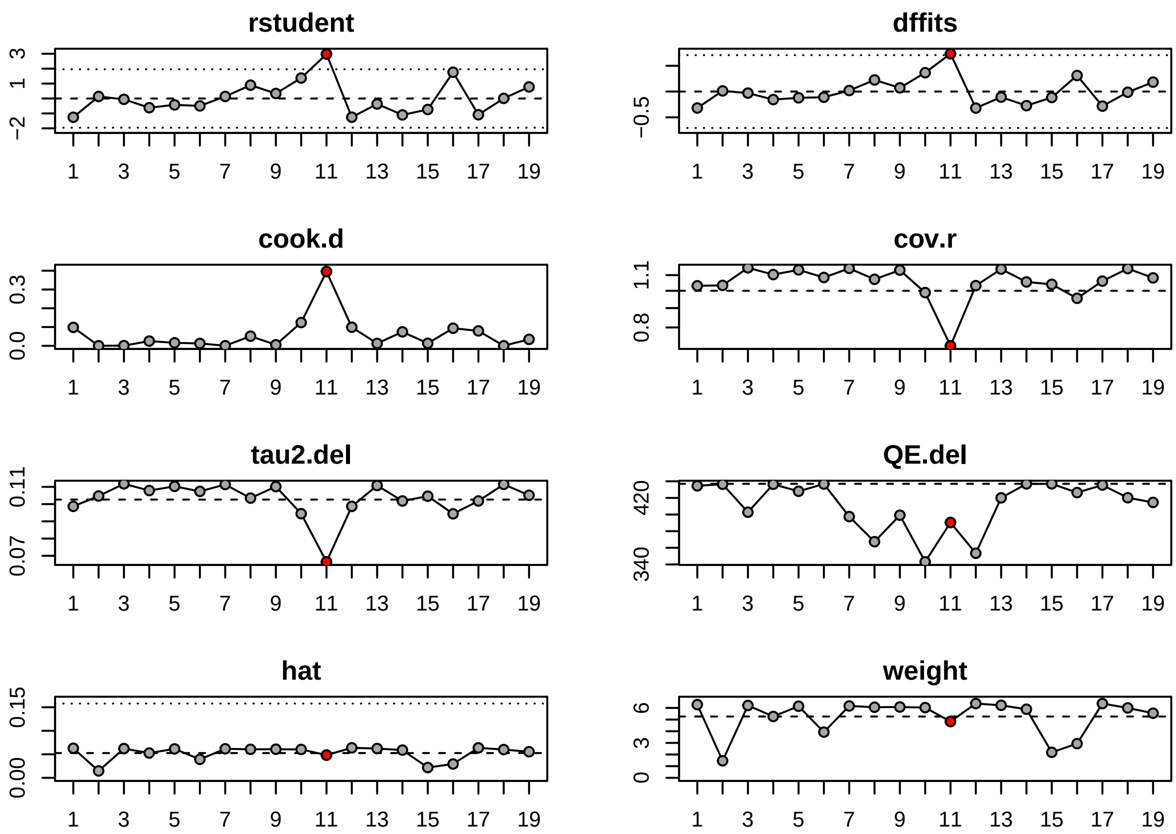
```
## The following objects are masked from 'package:metamisc':
##
##     forest, se
```

```r
# attach real study labels so plots use them instead of row numbers
rma_meta <- rma( # random effect meta analysis
  yi = logit_c, #observed effect
  sei = se_logit, # each studies precision and weight
```

```r
  data = df_primary_logit, # location of variables
         method = "REML", # methos for calculating study estimate
  slab = df_primary_logit$Index)


inf <- influence(rma_meta) # calcualte the influence statistic for every study

print(inf$inf)
```

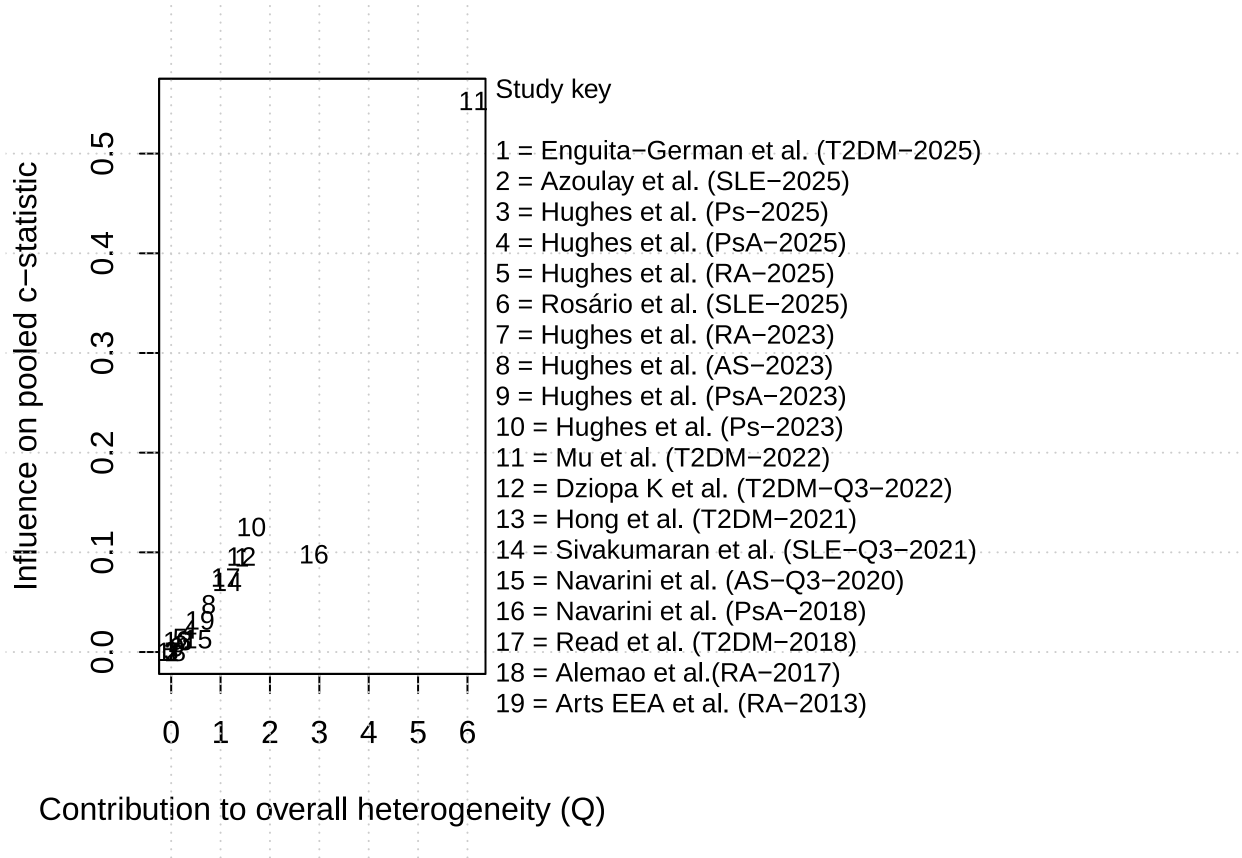
```
##
##     rstudent  dffits cook.d  cov.r tau2.del   QE.del    hat weight inf
## 1    -1.2585 -0.3193 0.0981 1.0308   0.0987 434.5363 0.0628 6.2832
## 3     0.1341  0.0127 0.0002 1.0333   0.1047 436.4591 0.0146 1.4606
## 6a   -0.0561 -0.0298 0.0010 1.1499   0.1118 402.8114 0.0621 6.2105
## 6b   -0.6181 -0.1555 0.0252 1.1037   0.1079 436.4052 0.0527 5.2678
## 6c   -0.4259 -0.1216 0.0158 1.1357   0.1103 427.9982 0.0615 6.1457
## 7    -0.4979 -0.1105 0.0125 1.0843   0.1074 436.7186 0.0392 3.9246
## 8a    0.1370  0.0200 0.0004 1.1459   0.1114 397.5324 0.0617 6.1684
## 8b    0.8943  0.2255 0.0512 1.0724   0.1035 367.3391 0.0606 6.0617
## 8c    0.3361  0.0721 0.0056 1.1340   0.1102 399.1945 0.0608 6.0794
## 8d    1.3712  0.3662 0.1240 0.9889   0.0944 343.1523 0.0603 6.0311
## 16    2.9803  0.7344 0.3956 0.7154   0.0665 390.4343 0.0483 4.8335   *
## 19b  -1.2540 -0.3209 0.0990 1.0319   0.0987 353.4536 0.0638 6.3760
## 23   -0.3673 -0.1080 0.0126 1.1410   0.1108 420.0769 0.0623 6.2254
## 26b  -1.1006 -0.2738 0.0744 1.0552   0.1018 436.8518 0.0589 5.8887
## 29b  -0.7371 -0.1151 0.0133 1.0400   0.1046 436.9280 0.0219 2.1882
## 35    1.7602  0.3118 0.0936 0.9553   0.0944 426.3958 0.0293 2.9346
## 36   -1.0895 -0.2830 0.0794 1.0606   0.1018 435.4247 0.0637 6.3729
## 39    0.0076 -0.0131 0.0002 1.1447   0.1114 420.2201 0.0600 6.0009
## 40    0.7749  0.1833 0.0343 1.0817   0.1051 414.5785 0.0555 5.5469
```

```r
plot(inf) # plot influence diagnostics
```




```r
# x-axis = each study's contribution to overall heterogeneity
# y-axis = its influence on the pooled estimate
# Build the study key from the model's data, in row order (Baujat labels by row position)
study_key <- paste0(
  seq_len(nrow(df_primary_logit)), # extract the number of rows from the dataframe
  " = ",
  df_primary_logit$Author) # author name would be combined with row number

plot_settings <- par( # save the graphic setting inside
  mar = c(5, 4, 2, 20), # control lot margins to create better spacing
  xpd = NA) # allow text outside normal plotting region

baujat( # create baugat plot
  rma_meta,
  xlab = "Contribution to overall heterogeneity (Q)",
  ylab = "Influence on pooled c-statistic",
  main = ""
)

usr <- par("usr") # place study key outside the graph for identificarion

text(
  usr[2] + 0.2,
  usr[4],
  labels = paste(
    c("Study key", "", study_key),
```

```
    collapse = "\n"),
  adj = c(0, 1),
  cex = 0.8
)
```




```
par(plot_settings)
```

```
# check difference in two dataframe
library(dplyr)
df_all_only <- anti_join( # keep only rows in first table that do not have a match in the
↪  second table
  df_all, # searching table
  df_primary, # reference table
  by = 'Author') # compare the two dataframe using the Author column
print(df_all_only) # print result
```

```
## # A tibble: 15 x 46
##    Number Index Author   Title Sample_size Study_Year Disease_Group QRISK_Version
##     <dbl> <chr> <chr>    <chr>       <dbl>      <dbl> <chr>         <chr>
## 1       4 4     Krishn~  04 C~        5371       2024 "Diabetes (T~ QRISK3
## 2       5 5a    Llorca~  05 P~         708       2012 "\r\n Rheuma~ QRISK3
## 3       5 5b    Llorca~  05 P~         692       2012 "Ankylosing ~ QRISK3
## 4       5 5c    Llorca~  05 P~         680       2012 "Psoriatic A~ QRISK3
## 5      14 14    Panopo~  14 G~         124       2023 "Systemic Lu~ QRISK3
```

```
## 6      18 18    Queved~ 18 {~         296       2022 "Systemic Lu~ QRISK3
## 7      19 19a   Dziopa~ 19 C~      168871       2022 "Diabetes (T~ QRISK2
## 8      21 21    Nedogo~ 21 I~         750       2021 "Metabolic S~ QRISK3
## 9      22 22    Carril~ 22 Q~         186       2021 "Inflammator~ QRISK3
## 10     26 26a   Sivaku~ 26 A~        1887       2021 "Systemic Lu~ QRISK2
## 11     27 27b   Drosos~ 27 U~         210       2020 "Systemic Lu~ QRISK3
## 12     28 28    Gerasi~ 28 A~         109       2021 "Rheumatoid ~ QRISK3
## 13     29 29a   Navari~ 29 C~         133       2020 "Ankylosing ~ QRISK2
## 14     32 32    Viswan~ 32 I~         339       2020 "chronic kid~ QRISK3
## 15     37 37    Shen e~ 37 U~         146       2018 "Psoriatic A~ QRISK2
## # i 38 more variables: Outcome <chr>, AUC <chr>, Calibration <chr>,
## #   `Follow-Up` <chr>, Country <chr>, Participants <chr>, Predictors2 <chr>,
## #   Outcome3 <chr>, Analysis <chr>, Overall <chr>, Metanalysis <dbl>,
## #   Confidence_interval <chr>, Dscrimination <chr>, QRISK_2 <dbl>,
## #   QRISK_3 <dbl>, Outcome_type <chr>, Outcome_Description <chr>,
## #   Applicability <dbl>, Include <dbl>, PA <chr>, PR <chr>, OT <chr>, OA <chr>,
## #   Study_2 <chr>, `Single_vs_multi-centre` <chr>, Events <dbl>, ...
```

SUBGROUP ANALYSIS - Disease group

```r
library(meta)

primary_subgroup_disease <- metagen(
  TE               = logit_c, # use logit estimates
  seTE             = se_logit, # use logit SE
  studlab          = Author, # identify by author
  data             = df_primary_logit, # find all in this dataframe
  sm               = "PLOGIT", # inform that the estimate are in logit
  method.tau       = "REML", # how to calculate the between study variance
  method.random.ci = "HK", # how to calcualte the CI
  subgroup         = Disease_Group, # split the studies according to disease group
  tau.common       = TRUE, # make all studeis in the group share the same tau square by
   ↪ pooling the estimates as few numbers of study would make seperate tau unstable
  prediction       = TRUE # calculate the prediction interval for each group
)

summary(primary_subgroup_disease)
```

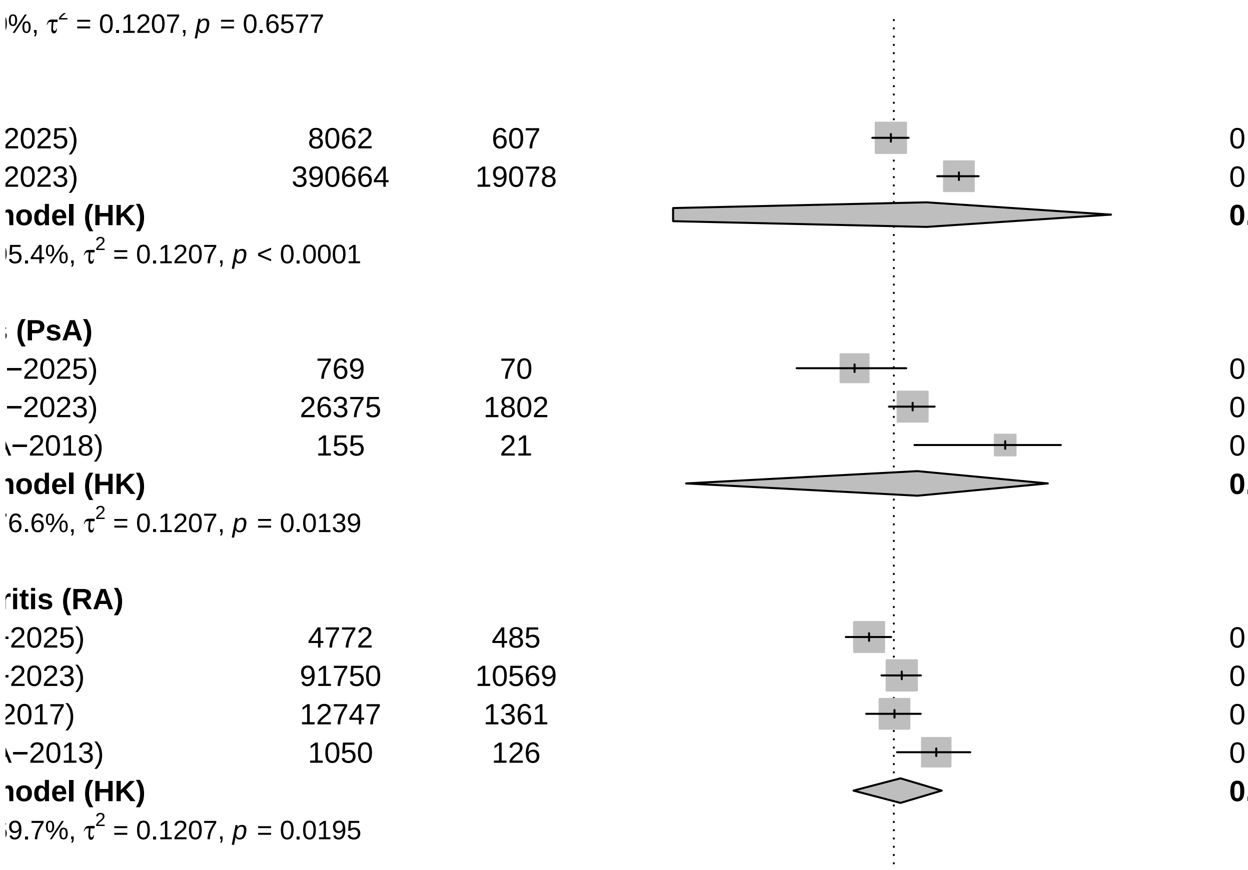
```
##                                    proportion           95% CI %W(common)
## Enguita-German et al. (T2DM-2025)      0.6630 [0.6453; 0.6803]        2.8
## Azoulay et al. (SLE-2025)              0.7600 [0.5001; 0.9093]        0.0
## Hughes et al. (Ps-2025)                0.7400 [0.7195; 0.7595]        1.6
## Hughes et al. (PsA-2025)               0.7000 [0.6361; 0.7569]        0.2
## Hughes et al. (RA-2025)                0.7160 [0.6904; 0.7403]        1.1
## Rosário et al. (SLE-2025)              0.7030 [0.5902; 0.7955]        0.1
## Hughes et al. (RA-2023)                0.7520 [0.7296; 0.7731]        1.3
## Hughes et al. (AS-2023)                0.7940 [0.7694; 0.8166]        0.8
## Hughes et al. (PsA-2023)               0.7640 [0.7379; 0.7883]        0.9
## Hughes et al. (Ps-2023)                0.8150 [0.7911; 0.8367]        0.8
## Mu et al. (T2DM-2022)                  0.8780 [0.8346; 0.9112]        0.1
## Dziopa K et al. (T2DM-Q3-2022)         0.6640 [0.6600; 0.6680]       53.7
## Hong et al. (T2DM-2021)                0.7200 [0.6995; 0.7396]        1.7
```

```
## Sivakumaran et al. (SLE-Q3-2021)      0.6700 [0.6287; 0.7088]        0.5
## Navarini et al. (AS-Q3-2020)          0.6600 [0.4487; 0.8224]        0.0
## Navarini et al. (PsA-2018)            0.8660 [0.7660; 0.9273]        0.0
## Read et al. (T2DM-2018)               0.6740 [0.6690; 0.6790]       33.3
## Alemao et al.(RA-2017)                0.7440 [0.7128; 0.7729]        0.7
## Arts EEA et al. (RA-2013)             0.7900 [0.7468; 0.8276]        0.3
##                                   %W(random)                    Disease_Group
## Enguita-German et al. (T2DM-2025)       6.3                  Diabetes (T2DM)
## Azoulay et al. (SLE-2025)               1.5 Systemic Lupus Erythematosus (SLE)
## Hughes et al. (Ps-2025)                 6.2                        Psoriasis
## Hughes et al. (PsA-2025)                5.3          Psoriatic Arthritis (PsA)
## Hughes et al. (RA-2025)                 6.1          Rheumatoid Arthritis (RA)
## Rosário et al. (SLE-2025)               3.9 Systemic Lupus Erythematosus (SLE)
## Hughes et al. (RA-2023)                 6.2          Rheumatoid Arthritis (RA)
## Hughes et al. (AS-2023)                 6.1        Ankylosing Spondylitis (AS)
## Hughes et al. (PsA-2023)                6.1          Psoriatic Arthritis (PsA)
## Hughes et al. (Ps-2023)                 6.0                        Psoriasis
## Mu et al. (T2DM-2022)                   4.8                  Diabetes (T2DM)
## Dziopa K et al. (T2DM-Q3-2022)          6.4                  Diabetes (T2DM)
## Hong et al. (T2DM-2021)                 6.2                  Diabetes (T2DM)
## Sivakumaran et al. (SLE-Q3-2021)        5.9 Systemic Lupus Erythematosus (SLE)
## Navarini et al. (AS-Q3-2020)            2.2        Ankylosing Spondylitis (AS)
## Navarini et al. (PsA-2018)              2.9          Psoriatic Arthritis (PsA)
## Read et al. (T2DM-2018)                 6.4                  Diabetes (T2DM)
## Alemao et al.(RA-2017)                  6.0          Rheumatoid Arthritis (RA)
## Arts EEA et al. (RA-2013)               5.5          Rheumatoid Arthritis (RA)
##
## Number of studies: k = 19
##
##                            proportion           95% CI    z|t  p-value
## Common effect model            0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)      0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                   [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1207 [0.0537; 0.4082]; tau = 0.3474 [0.2318; 0.6389]
##  I^2 = 89.6% [98.0%; 99.7%]; H = 3.10 [6.99; 19.09]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Results for subgroups (common effect model):
##                                                        k proportion
## Disease_Group = Diabetes (T2DM)                        5     0.6692
## Disease_Group = Systemic Lupus Erythematosus (SLE)   3     0.6760
## Disease_Group = Psoriasis                              2     0.7661
## Disease_Group = Psoriatic Arthritis (PsA)              3     0.7572
## Disease_Group = Rheumatoid Arthritis (RA)              4     0.7421
## Disease_Group = Ankylosing Spondylitis (AS)            2     0.7910
```

```
##                                                             95% CI     Q   I^2
## Disease_Group = Diabetes (T2DM)                    [0.6661; 0.6722] 81.96 95.1%
## Disease_Group = Systemic Lupus Erythematosus (SLE) [0.6380; 0.7118]  0.84  0.0%
## Disease_Group = Psoriasis                          [0.7504; 0.7811] 21.74 95.4%
## Disease_Group = Psoriatic Arthritis (PsA)          [0.7337; 0.7792]  8.56 76.6%
## Disease_Group = Rheumatoid Arthritis (RA)          [0.7282; 0.7555]  9.89 69.7%
## Disease_Group = Ankylosing Spondylitis (AS)        [0.7665; 0.8135]  2.33 57.1%
##
## Test for subgroup differences (common effect model):
##                     Q d.f.  p-value
## Between groups 311.65    5 < 0.0001
## Within groups  125.31   13 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                                                     k proportion
## Disease_Group = Diabetes (T2DM)                     5     0.7227
## Disease_Group = Systemic Lupus Erythematosus (SLE)  3     0.6953
## Disease_Group = Psoriasis                           2     0.7793
## Disease_Group = Psoriatic Arthritis (PsA)           3     0.7693
## Disease_Group = Rheumatoid Arthritis (RA)           4     0.7506
## Disease_Group = Ankylosing Spondylitis (AS)         2     0.7603
##                                                             95% CI  tau^2
## Disease_Group = Diabetes (T2DM)                    [0.5781; 0.8321] 0.1207
## Disease_Group = Systemic Lupus Erythematosus (SLE) [0.5912; 0.7825] 0.1207
## Disease_Group = Psoriasis                          [0.1804; 0.9827] 0.1207
## Disease_Group = Psoriatic Arthritis (PsA)          [0.5143; 0.9130] 0.1207
## Disease_Group = Rheumatoid Arthritis (RA)          [0.6988; 0.7961] 0.1207
## Disease_Group = Ankylosing Spondylitis (AS)        [0.0586; 0.9939] 0.1207
##                                                       tau
## Disease_Group = Diabetes (T2DM)                    0.3474
## Disease_Group = Systemic Lupus Erythematosus (SLE) 0.3474
## Disease_Group = Psoriasis                          0.3474
## Disease_Group = Psoriatic Arthritis (PsA)          0.3474
## Disease_Group = Rheumatoid Arthritis (RA)          0.3474
## Disease_Group = Ankylosing Spondylitis (AS)        0.3474
##
## Test for subgroup differences (random effects model (HK)):
##                     Q d.f.  p-value
## Between groups   6.34    5   0.2748
## Within groups  125.31   13 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

FORREST PLOT - SUBGROUP ANALYSIS - Disease group

```r
meta::forest(
  primary_subgroup_disease,
  backtransf = TRUE,
  xlab = "c-statistic",
  xlim = c(0.5, 1),
  at = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
  leftcols  = c("studlab", "Sample_size", "Events"),
  leftlabs  = c("Study", "N", "CVD events"),
  rightcols = c("effect", "ci"),
  rightlabs = c("c-statistic", "95% CI"),
  common = FALSE,
  test.subgroup = TRUE,
  print.subgroup.name = FALSE,
  addrows.below.overall = 5,
  colgap = "8mm",
  colgap.left = "6mm",
  colgap.forest.left = "10mm",
  fontsize = 11
)
```




```r
# Between-group difference test
primary_subgroup_disease$Q.b.random      # Q between random effect
```

```
## [1] 6.337486
```

```r
primary_subgroup_disease$df.Q.b          # degree of freedom
```

```
## [1] 5
```

```r
primary_subgroup_disease$pval.Q.b.random # p-value
```

```
## [1] 0.2747509
```

```r
# Per-subgroup back-transformed estimates
data.frame(
  disease = primary_subgroup_disease$subgroup.levels,
  k       = primary_subgroup_disease$k.w,
  c_stat  = plogis(primary_subgroup_disease$TE.random.w),
  lower   = plogis(primary_subgroup_disease$lower.random.w),
  upper   = plogis(primary_subgroup_disease$upper.random.w),
  I2      = primary_subgroup_disease$I2.w
)
```

```
##                                                                 disease k
## Diabetes (T2DM)                                         Diabetes (T2DM) 5
## Systemic Lupus Erythematosus (SLE) Systemic Lupus Erythematosus (SLE) 3
## Psoriasis                                                     Psoriasis 2
## Psoriatic Arthritis (PsA)                       Psoriatic Arthritis (PsA) 3
## Rheumatoid Arthritis (RA)                       Rheumatoid Arthritis (RA) 4
## Ankylosing Spondylitis (AS)                   Ankylosing Spondylitis (AS) 2
##                                       c_stat      lower     upper        I2
## Diabetes (T2DM)                    0.7226928 0.57806361 0.8321424 0.9511940
## Systemic Lupus Erythematosus (SLE) 0.6952538 0.59123248 0.7825404 0.0000000
## Psoriasis                          0.7793119 0.18041171 0.9826538 0.9539983
## Psoriatic Arthritis (PsA)          0.7692962 0.51433085 0.9130409 0.7663248
## Rheumatoid Arthritis (RA)          0.7506201 0.69882402 0.7961073 0.6965992
## Ankylosing Spondylitis (AS)        0.7602706 0.05855809 0.9938536 0.5705511
```

SUBGROUP ANALYSIS BY QRISK VERSION

```r
# See version information
table(df_primary_logit$QRISK_Version, useNA = "ifany")
```

```
##
## QRISK2 QRISK3
##      5     14
```

```r
# subgroup analysis QRISK version
library(meta)

primary_subgroup_version <- metagen(
  TE                = logit_c,
  seTE              = se_logit,
```

```r
  studlab          = Author,
  data             = df_primary_logit,
  sm               = "PLOGIT",
  method.tau       = "REML",
  method.random.ci = "HK",
  subgroup         = QRISK_Version,
  tau.common       = TRUE,
  prediction       = TRUE
)

summary(primary_subgroup_version)
```

```
##                                   proportion           95% CI %W(common)
## Enguita-German et al. (T2DM-2025)     0.6630 [0.6453; 0.6803]        2.8
## Azoulay et al. (SLE-2025)             0.7600 [0.5001; 0.9093]        0.0
## Hughes et al. (Ps-2025)               0.7400 [0.7195; 0.7595]        1.6
## Hughes et al. (PsA-2025)              0.7000 [0.6361; 0.7569]        0.2
## Hughes et al. (RA-2025)               0.7160 [0.6904; 0.7403]        1.1
## Rosário et al. (SLE-2025)             0.7030 [0.5902; 0.7955]        0.1
## Hughes et al. (RA-2023)               0.7520 [0.7296; 0.7731]        1.3
## Hughes et al. (AS-2023)               0.7940 [0.7694; 0.8166]        0.8
## Hughes et al. (PsA-2023)              0.7640 [0.7379; 0.7883]        0.9
## Hughes et al. (Ps-2023)               0.8150 [0.7911; 0.8367]        0.8
## Mu et al. (T2DM-2022)                 0.8780 [0.8346; 0.9112]        0.1
## Dziopa K et al. (T2DM-Q3-2022)        0.6640 [0.6600; 0.6680]       53.7
## Hong et al. (T2DM-2021)               0.7200 [0.6995; 0.7396]        1.7
## Sivakumaran et al. (SLE-Q3-2021)      0.6700 [0.6287; 0.7088]        0.5
## Navarini et al. (AS-Q3-2020)          0.6600 [0.4487; 0.8224]        0.0
## Navarini et al. (PsA-2018)            0.8660 [0.7660; 0.9273]        0.0
## Read et al. (T2DM-2018)               0.6740 [0.6690; 0.6790]       33.3
## Alemao et al.(RA-2017)                0.7440 [0.7128; 0.7729]        0.7
## Arts EEA et al. (RA-2013)             0.7900 [0.7468; 0.8276]        0.3
##                                   %W(random) QRISK_Version
## Enguita-German et al. (T2DM-2025)        6.3        QRISK2
## Azoulay et al. (SLE-2025)                1.5        QRISK3
## Hughes et al. (Ps-2025)                  6.2        QRISK3
## Hughes et al. (PsA-2025)                 5.3        QRISK3
## Hughes et al. (RA-2025)                  6.1        QRISK3
## Rosário et al. (SLE-2025)                3.9        QRISK3
## Hughes et al. (RA-2023)                  6.2        QRISK3
## Hughes et al. (AS-2023)                  6.1        QRISK3
## Hughes et al. (PsA-2023)                 6.1        QRISK3
## Hughes et al. (Ps-2023)                  6.0        QRISK3
## Mu et al. (T2DM-2022)                    4.8        QRISK3
## Dziopa K et al. (T2DM-Q3-2022)           6.4        QRISK3
## Hong et al. (T2DM-2021)                  6.2        QRISK3
## Sivakumaran et al. (SLE-Q3-2021)         5.9        QRISK3
## Navarini et al. (AS-Q3-2020)             2.2        QRISK3
## Navarini et al. (PsA-2018)               2.9        QRISK2
## Read et al. (T2DM-2018)                  6.4        QRISK2
## Alemao et al.(RA-2017)                   6.0        QRISK2
## Arts EEA et al. (RA-2013)                5.5        QRISK2
##
```

```
## Number of studies: k = 19
##
##                           proportion           95% CI    z|t  p-value
## Common effect model           0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)     0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                  [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1097 [0.0539; 0.3012]; tau = 0.3312 [0.2322; 0.5488]
##  I^2 = 96.1% [94.4%; 98.9%]; H = 5.07 [4.23; 9.76]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Results for subgroups (common effect model):
##                          k proportion           95% CI      Q   I^2
## QRISK_Version = QRISK2   5     0.6758 [0.6711; 0.6806]  52.77 92.4%
## QRISK_Version = QRISK3  14     0.6770 [0.6734; 0.6806] 384.04 96.6%
##
## Test for subgroup differences (common effect model):
##                      Q d.f.  p-value
## Between groups   0.14    1   0.7035
## Within groups  436.81   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                          k proportion           95% CI  tau^2    tau
## QRISK_Version = QRISK2   5     0.7387 [0.6245; 0.8278] 0.1097 0.3312
## QRISK_Version = QRISK3  14     0.7452 [0.7049; 0.7817] 0.1097 0.3312
##
## Test for subgroup differences (random effects model (HK)):
##                      Q d.f.  p-value
## Between groups   0.03    1   0.8735
## Within groups  436.81   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

```r
# Extract the between-group test and per-version estimate
# Between-group difference
primary_subgroup_version$Q.b.random
```

```
## [1] 0.02534123
```

```r
primary_subgroup_version$df.Q.b
```

```
## [1] 1
```

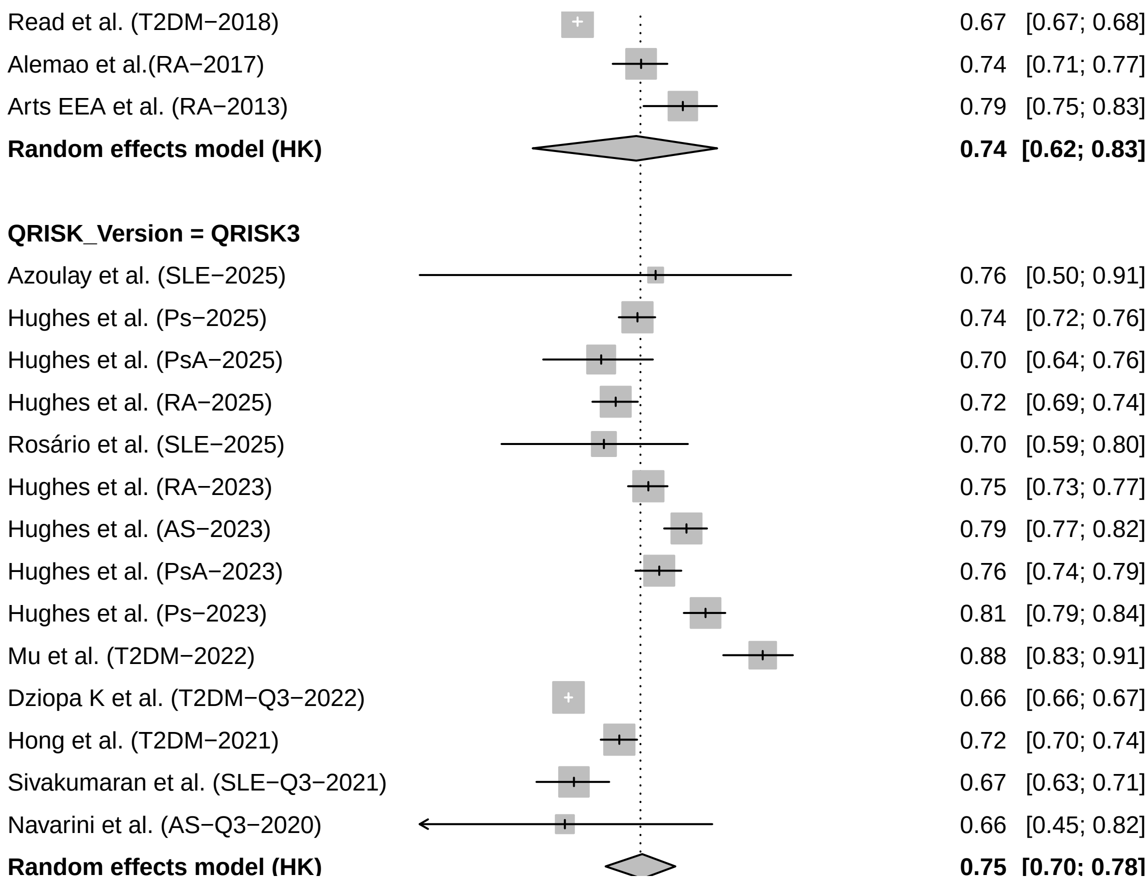
```r
primary_subgroup_version$pval.Q.b.random
```

```
## [1] 0.8735197
```

```r
# Per-version back-transformed estimates
data.frame(
  version = primary_subgroup_version$subgroup.levels,
  k       = primary_subgroup_version$k.w,
  c_stat  = plogis(primary_subgroup_version$TE.random.w),
  lower   = plogis(primary_subgroup_version$lower.random.w),
  upper   = plogis(primary_subgroup_version$upper.random.w),
  I2      = primary_subgroup_version$I2.w
)
```

```
##        version  k    c_stat     lower     upper        I2
## QRISK2  QRISK2  5 0.7387317 0.6245366 0.8277729 0.9242035
## QRISK3  QRISK3 14 0.7452183 0.7048955 0.7817374 0.9661492
```

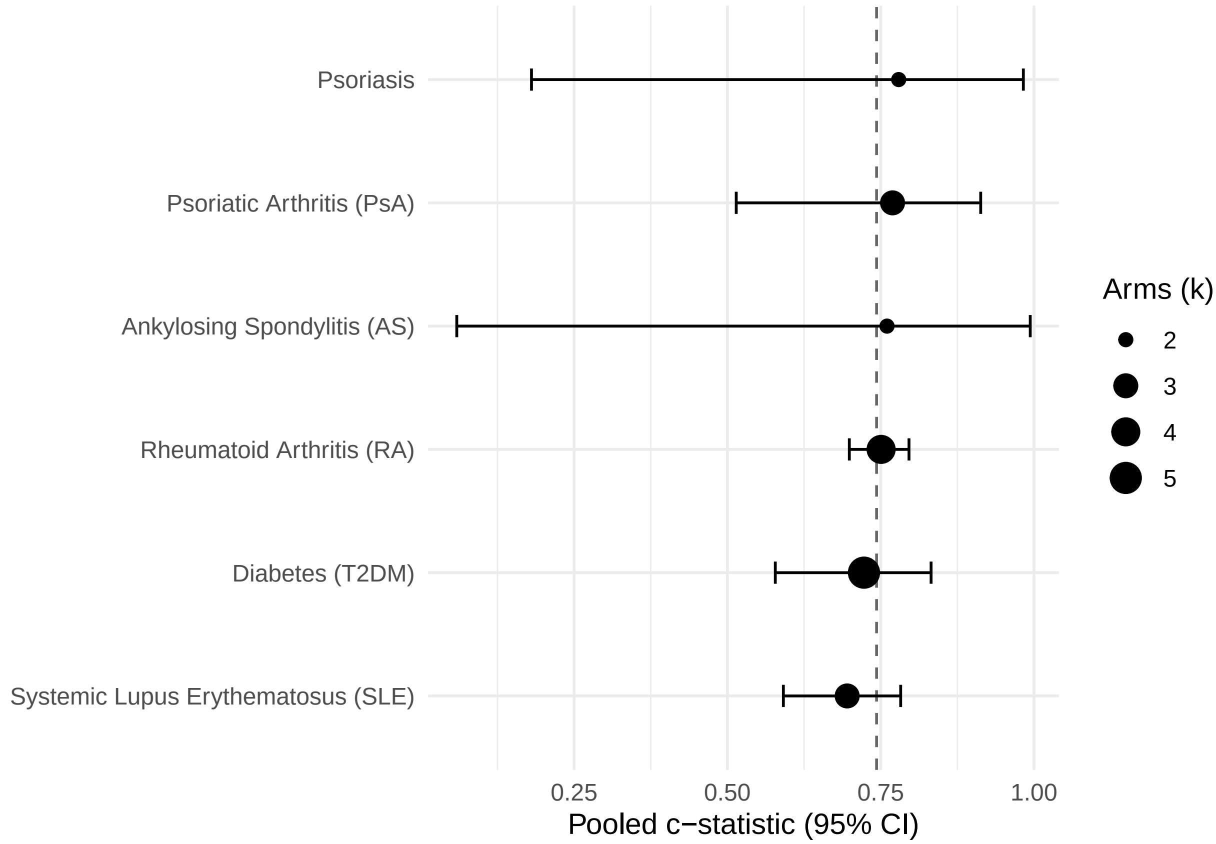
```r
# widen the bottom margin before plotting
par(mar = c(8, 4, 2, 2))   # bottom, left, top, right - more bottom space

meta::forest(primary_subgroup_version,
             backtransf = TRUE,
             xlab       = "c-statistic",
             xlim       = c(0.5, 1),
             at         = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),
             leftcols   = "studlab", leftlabs = "Study",
             rightcols  = c("effect", "ci"), rightlabs = c("c-statistic", "95% CI"),
             common          = FALSE,
             hetstat         = FALSE,
             test.subgroup   = TRUE,
             overall.hetstat = TRUE,
             fontsize        = 9,
             spacing         = 1.1)
```




```r
# subgroup summary
library(ggplot2)
sg <- data.frame(
  group = primary_subgroup_disease$subgroup.levels,
  k     = primary_subgroup_disease$k.w,
  est   = plogis(primary_subgroup_disease$TE.random.w),
  lo    = plogis(primary_subgroup_disease$lower.random.w),
  hi    = plogis(primary_subgroup_disease$upper.random.w))

ggplot(sg, aes(x = est, y = reorder(group, est))) +
  geom_vline(xintercept = plogis(primary_logit$TE.random),
             linetype = 2, colour = "grey40") +
  geom_errorbarh(aes(xmin = lo, xmax = hi), height = 0.18) +
  geom_point(aes(size = k)) +
  scale_size_continuous(range = c(2, 5), name = "Arms (k)") +
  labs(x = "Pooled c-statistic (95% CI)", y = NULL) +
  theme_minimal(base_size = 11)
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `height` was translated to `width`.
```



PAIRED QRISK VERSION COMPARISM

```r
library(dplyr)
library(stringr)
library(tidyr)
```

```
##
## Attaching package: 'tidyr'

## The following objects are masked from 'package:Matrix':
##
##     expand, pack, unpack
```

```r
# Extract the six paired arms from df_all
paired <- df_all %>% # create dataset of paired studies
  filter(Index %in% c("19a", "19b", "26a", "26b", "29a", "29b")) %>% # select the paired
  ↪ studeis
  mutate( # add new columns to existing ones
    cohort  = str_extract(Author, "\\d+"),                        # 19, 26, 29
    version = ifelse(str_detect(Author, "a$"), "QRISK2", "QRISK3")
  )
```

```r
# VERIFY the a/b version mapping against the data's own version field
paired %>% select(Author, cohort, version, QRISK_Version, Estimate, LowerCI, UpperCI)
```

```
## # A tibble: 6 x 7
##   Author                   cohort version QRISK_Version Estimate LowerCI UpperCI
##   <chr>                    <chr>  <chr>   <chr>            <dbl>   <dbl>   <dbl>
## 1 Dziopa K et al. (T2DM-Q~ 2      QRISK3  QRISK2           0.664    0.66   0.667
## 2 Dziopa K et al. (T2DM-Q~ 2      QRISK3  QRISK3           0.664    0.66   0.668
## 3 Sivakumaran et al. (SLE~ 2      QRISK3  QRISK2           0.72     0.68   0.76
## 4 Sivakumaran et al. (SLE~ 3      QRISK3  QRISK3           0.67     0.63   0.71
## 5 Navarini et al. (AS-Q3-~ 3      QRISK3  QRISK3           0.66     0.48   0.84
## 6 Navarini et al. (AS-Q2-~ 2      QRISK3  QRISK2           0.68     0.5    0.86
```

```r
library(dplyr)
library(tidyr)
library(readr)

# Build the paired set
paired <- df_all %>%
  filter(Index %in% c("19a","19b","26a","26b","29a","29b")) %>%
  mutate(
    Index   = trimws(Index), # trim whitespace from the index
    cohort  = parse_number(Index), # extract the number part from the index
    version = QRISK_Version # store the qrisk version in version
  )

# verify 6 rows, each cohort has one QRISK2 and one QRISK3
print(count(paired, cohort, version))
```

```
## # A tibble: 6 x 3
##   cohort version     n
##    <dbl> <chr>   <int>
## 1     19 QRISK2      1
## 2     19 QRISK3      1
## 3     26 QRISK2      1
## 4     26 QRISK3      1
## 5     29 QRISK2      1
## 6     29 QRISK3      1
```

```r
# Logit-transform each arm c-statistic
paired <- paired %>%
  mutate( # create column for each new logit transformed variable
    logit_c  = log(Estimate / (1 - Estimate)),
    logit_lb = log(LowerCI  / (1 - LowerCI)),
    logit_ub = log(UpperCI  / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96)
  )

# Reshape to one row per cohort, QRISK2 and QRISK3 side by side
wide <- paired %>%
  select(cohort, version, Estimate, logit_c, se_logit) %>%
  pivot_wider( # convert to data to wide form
```

```r
    id_cols     = cohort,
    names_from  = version,
    values_from = c(Estimate, logit_c, se_logit) # variables to be widened
  )

print(names(wide))
```

```
## [1] "cohort"          "Estimate_QRISK2" "Estimate_QRISK3" "logit_c_QRISK2"
## [5] "logit_c_QRISK3"  "se_logit_QRISK2" "se_logit_QRISK3"
```

```r
# Within-cohort difference (QRISK3 - QRISK2)
wide <- wide %>%
  mutate(
    delta_logit    = logit_c_QRISK3 - logit_c_QRISK2,
    delta_cstat    = Estimate_QRISK3 - Estimate_QRISK2,
    se_delta_indep = sqrt(se_logit_QRISK3^2 + se_logit_QRISK2^2)   # SE assuming
     ↪  independence
  )

# Present the paired comparison
wide %>%
  transmute(
    Cohort                   = cohort,
    `QRISK2 c`               = round(Estimate_QRISK2, 3),
    `QRISK3 c`               = round(Estimate_QRISK3, 3),
    `Delta c (3-2)`          = round(delta_cstat, 3),
    `Delta logit (3-2)`      = round(delta_logit, 3),
    `SE (indep, overest)` = round(se_delta_indep, 3)
  )
```

```
## # A tibble: 3 x 6
##   Cohort `QRISK2 c` `QRISK3 c` `Delta c (3-2)` `Delta logit (3-2)`
##    <dbl>      <dbl>      <dbl>           <dbl>               <dbl>
## 1     19      0.664      0.664            0                   0
## 2     26      0.72       0.67            -0.05               -0.236
## 3     29      0.68       0.66            -0.02               -0.09
## # i 1 more variable: `SE (indep, overest)` <dbl>
```

```r
# check the average improvement of discrimination between 3 and 2
mean_delta_logit <- mean(wide$delta_logit)
cat("Mean within-cohort logit difference (QRISK3 - QRISK2):",
    round(mean_delta_logit, 4), "\n")
```

```
## Mean within-cohort logit difference (QRISK3 - QRISK2): -0.1089
```

SUBGROUP ANALYSIS (PROBAST)

```r
# Which column holds the overall PROBAST ROB verdict, and how is it coded?
table(df_primary_logit$Overall, useNA = "ifany")
```

```
##
## HIGH  LOW
##   11    8
```

```r
# ROB SUBGROUP ANALYSIS
library(meta)

primary_subgroup_rob <- metagen(
  TE                 = logit_c,
  seTE               = se_logit,
  studlab            = Author,
  data               = df_primary_logit,
  sm                 = "PLOGIT",
  method.tau         = "REML",
  method.random.ci   = "HK",
  subgroup           = Overall,
  tau.common         = TRUE,
  prediction         = TRUE
)

summary(primary_subgroup_rob)
```

```
##                                    proportion           95% CI %W(common)
## Enguita-German et al. (T2DM-2025)      0.6630 [0.6453; 0.6803]        2.8
## Azoulay et al. (SLE-2025)              0.7600 [0.5001; 0.9093]        0.0
## Hughes et al. (Ps-2025)                0.7400 [0.7195; 0.7595]        1.6
## Hughes et al. (PsA-2025)               0.7000 [0.6361; 0.7569]        0.2
## Hughes et al. (RA-2025)                0.7160 [0.6904; 0.7403]        1.1
## Rosário et al. (SLE-2025)              0.7030 [0.5902; 0.7955]        0.1
## Hughes et al. (RA-2023)                0.7520 [0.7296; 0.7731]        1.3
## Hughes et al. (AS-2023)                0.7940 [0.7694; 0.8166]        0.8
## Hughes et al. (PsA-2023)               0.7640 [0.7379; 0.7883]        0.9
## Hughes et al. (Ps-2023)                0.8150 [0.7911; 0.8367]        0.8
## Mu et al. (T2DM-2022)                  0.8780 [0.8346; 0.9112]        0.1
## Dziopa K et al. (T2DM-Q3-2022)         0.6640 [0.6600; 0.6680]       53.7
## Hong et al. (T2DM-2021)                0.7200 [0.6995; 0.7396]        1.7
## Sivakumaran et al. (SLE-Q3-2021)       0.6700 [0.6287; 0.7088]        0.5
## Navarini et al. (AS-Q3-2020)           0.6600 [0.4487; 0.8224]        0.0
## Navarini et al. (PsA-2018)             0.8660 [0.7660; 0.9273]        0.0
## Read et al. (T2DM-2018)                0.6740 [0.6690; 0.6790]       33.3
## Alemao et al.(RA-2017)                 0.7440 [0.7128; 0.7729]        0.7
## Arts EEA et al. (RA-2013)              0.7900 [0.7468; 0.8276]        0.3
##                                    %W(random) Overall
## Enguita-German et al. (T2DM-2025)         6.3    HIGH
## Azoulay et al. (SLE-2025)                 1.5    HIGH
## Hughes et al. (Ps-2025)                   6.2     LOW
## Hughes et al. (PsA-2025)                  5.3     LOW
## Hughes et al. (RA-2025)                   6.1     LOW
## Rosário et al. (SLE-2025)                 3.9    HIGH
## Hughes et al. (RA-2023)                   6.2     LOW
## Hughes et al. (AS-2023)                   6.1     LOW
## Hughes et al. (PsA-2023)                  6.1     LOW
## Hughes et al. (Ps-2023)                   6.0     LOW
## Mu et al. (T2DM-2022)                     4.8    HIGH
```

```
## Dziopa K et al. (T2DM-Q3-2022)          6.4     LOW
## Hong et al. (T2DM-2021)                 6.2    HIGH
## Sivakumaran et al. (SLE-Q3-2021)        5.9    HIGH
## Navarini et al. (AS-Q3-2020)            2.2    HIGH
## Navarini et al. (PsA-2018)              2.9    HIGH
## Read et al. (T2DM-2018)                 6.4    HIGH
## Alemao et al.(RA-2017)                  6.0    HIGH
## Arts EEA et al. (RA-2013)               5.5    HIGH
##
## Number of studies: k = 19
##
##                            proportion           95% CI    z|t  p-value
## Common effect model            0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)      0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                   [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1102 [0.0542; 0.3013]; tau = 0.3320 [0.2327; 0.5489]
##  I^2 = 96.1% [94.4%; 98.9%]; H = 5.06 [4.23; 9.75]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Results for subgroups (common effect model):
##                  k proportion           95% CI      Q   I^2
## Overall = HIGH  11     0.6788 [0.6742; 0.6833] 115.86 91.4%
## Overall = LOW    8     0.6751 [0.6714; 0.6788] 319.58 97.8%
##
## Test for subgroup differences (common effect model):
##                     Q d.f.  p-value
## Between groups   1.51    1   0.2187
## Within groups  435.44   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                  k proportion           95% CI  tau^2    tau
## Overall = HIGH  11     0.7411 [0.6796; 0.7944] 0.1102 0.3320
## Overall = LOW    8     0.7460 [0.7021; 0.7854] 0.1102 0.3320
##
## Test for subgroup differences (random effects model (HK)):
##                     Q d.f.  p-value
## Between groups   0.03    1   0.8742
## Within groups  435.44   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
```

```
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

```
# Between-group test and per-group estimates
# Between-group difference
primary_subgroup_rob$Q.b.random
```

```
## [1] 0.02506968
```

```
primary_subgroup_rob$df.Q.b
```

```
## [1] 1
```

```
primary_subgroup_rob$pval.Q.b.random
```

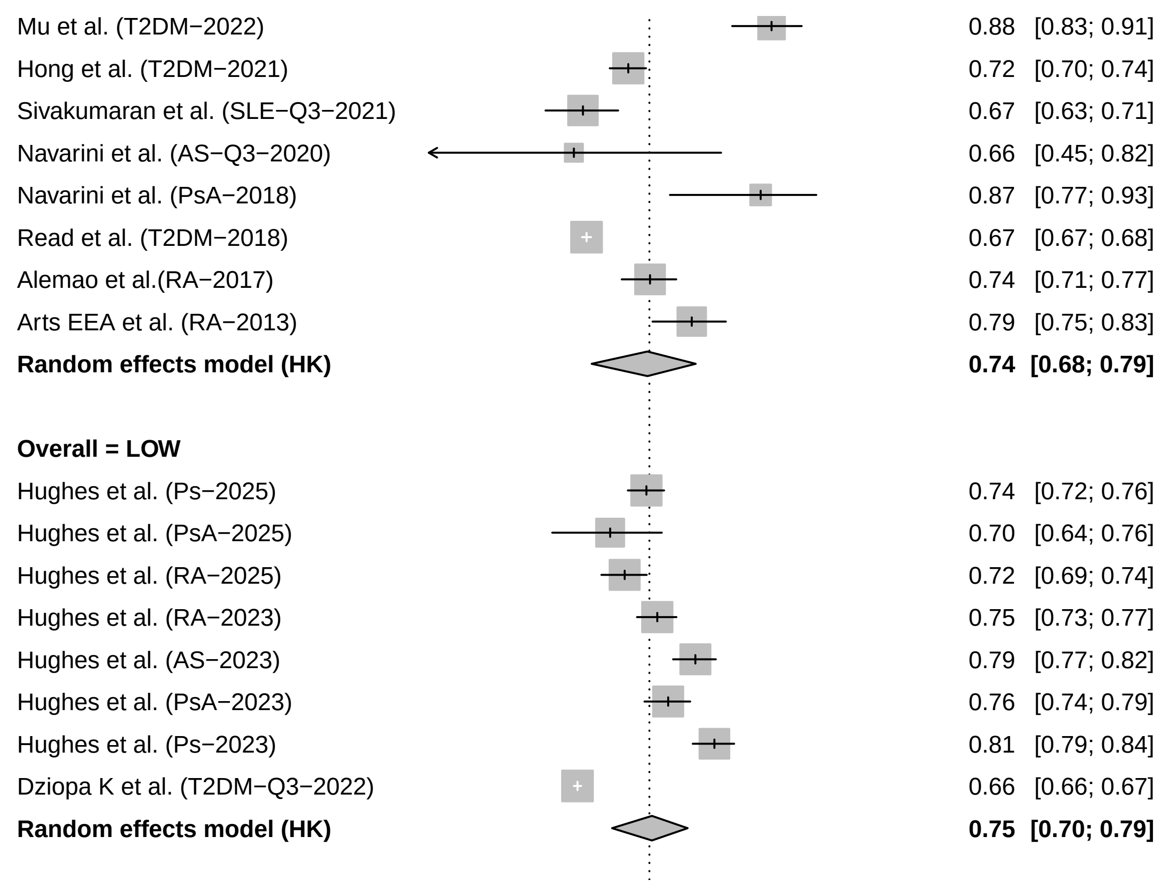
```
## [1] 0.8741936
```

```
# Per-ROB-group back-transformed estimates
data.frame(
  rob    = primary_subgroup_rob$subgroup.levels,
  k      = primary_subgroup_rob$k.w,
  c_stat = plogis(primary_subgroup_rob$TE.random.w),
  lower  = plogis(primary_subgroup_rob$lower.random.w),
  upper  = plogis(primary_subgroup_rob$upper.random.w),
  I2     = primary_subgroup_rob$I2.w
)
```

```
##       rob  k    c_stat     lower     upper        I2
## HIGH HIGH 11 0.7410989 0.6796174 0.7943521 0.9136900
## LOW   LOW  8 0.7460367 0.7021369 0.7854435 0.9780964
```

```
# FORST PLOT FOR ROB
par(mar = c(8, 4, 2, 2))

meta::forest(
  primary_subgroup_rob,
  backtransf = TRUE,
  xlab = "c-statistic",
  xlim = c(0.5, 1),
  at = c(0.5, 0.6, 0.7, 0.8, 0.9, 1.0),

  leftcols = "studlab",
  leftlabs = "Study",

  rightcols = c("effect", "ci"),
  rightlabs = c("c-statistic", "95% CI"),

  common = FALSE,
```

```
  hetstat = FALSE,
  overall.hetstat = TRUE,
  test.subgroup = TRUE,

  fontsize = 9,
  spacing = 1.1,

  addrows.below.overall = 2
)
```



SENSITIVITY ANALYSIS DROPING LOW STUDIES

```
df_lowrob <- df_primary_logit[df_primary_logit$Overall == "LOW", ]

primary_lowrob <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = df_lowrob,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  prediction = TRUE
```

```r
)
summary(primary_lowrob)
```

```
##      proportion           95% CI %W(common) %W(random)
## 6a       0.7400 [0.7195; 0.7595]        2.6       13.0
## 6b       0.7000 [0.6361; 0.7569]        0.3       10.2
## 6c       0.7160 [0.6904; 0.7403]        1.9       12.8
## 8a       0.7520 [0.7296; 0.7731]        2.1       12.9
## 8b       0.7940 [0.7694; 0.8166]        1.4       12.5
## 8c       0.7640 [0.7379; 0.7883]        1.5       12.6
## 8d       0.8150 [0.7911; 0.8367]        1.2       12.4
## 19b      0.6640 [0.6600; 0.6680]       88.9       13.6
##
## Number of studies: k = 8
##
##                             proportion           95% CI   z|t  p-value
## Common effect model             0.6751 [0.6714; 0.6788] 84.78        0
## Random effects model (HK)       0.7460 [0.7020; 0.7855] 11.56 < 0.0001
## Prediction interval                    [0.6072; 0.8481]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.0649 [0.0256; 0.2804]; tau = 0.2548 [0.1599; 0.5295]
##  I^2 = 97.8% [96.9%; 98.4%]; H = 6.76 [5.68; 8.03]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  319.58    7 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 7)
## - Prediction interval based on t-distribution (df = 7)
## - Logit transformation
```

SENSITIVITY ANALYSIS FOR DISCRIMINATION METRIC TYPE

```r
# CHECK TYPE
table(df_primary_logit$Dscrimination, useNA = "ifany")
```

```
##
##               AUC Harrell's C-index
##                16                 3
```

```r
# LEAVE OUT ONLY HARRELLS
library(meta)

# Keep only the AUC-metric studies (drop the three Harrell's C arms)
```

```r
df_auc_only <- df_primary_logit[df_primary_logit$Dscrimination == "AUC", ]
nrow(df_auc_only)
```

```
## [1] 16
```

```r
sens_auc_only <- metagen(
  TE               = logit_c,
  seTE             = se_logit,
  studlab          = Index,
  data             = df_auc_only,
  sm               = "PLOGIT",
  method.tau       = "REML",
  method.random.ci = "HK",
  prediction       = TRUE
)
summary(sens_auc_only)
```

```
##     proportion           95% CI %W(common) %W(random)
## 3       0.7600 [0.5001; 0.9093]        0.0        1.7
## 6a      0.7400 [0.7195; 0.7595]        1.7        7.3
## 6b      0.7000 [0.6361; 0.7569]        0.2        6.2
## 6c      0.7160 [0.6904; 0.7403]        1.2        7.3
## 7       0.7030 [0.5902; 0.7955]        0.1        4.6
## 8a      0.7520 [0.7296; 0.7731]        1.3        7.3
## 8b      0.7940 [0.7694; 0.8166]        0.9        7.2
## 8c      0.7640 [0.7379; 0.7883]        0.9        7.2
## 8d      0.8150 [0.7911; 0.8367]        0.8        7.1
## 16      0.8780 [0.8346; 0.9112]        0.1        5.7
## 19b     0.6640 [0.6600; 0.6680]       55.6        7.5
## 23      0.7200 [0.6995; 0.7396]        1.8        7.4
## 26b     0.6700 [0.6287; 0.7088]        0.5        7.0
## 29b     0.6600 [0.4487; 0.8224]        0.0        2.5
## 36      0.6740 [0.6690; 0.6790]       34.5        7.5
## 40      0.7900 [0.7468; 0.8276]        0.3        6.5
##
## Number of studies: k = 16
##
##                              proportion           95% CI    z|t  p-value
## Common effect model              0.6764 [0.6734; 0.6793] 108.02        0
## Random effects model (HK)        0.7434 [0.7069; 0.7767]  12.38 < 0.0001
## Prediction interval                     [0.5908; 0.8532]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.0992 [0.0454; 0.2641]; tau = 0.3149 [0.2132; 0.5139]
##  I^2 = 96.3% [95.1%; 97.2%]; H = 5.21 [4.53; 5.99]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  407.41   15 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
```

```
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 15)
## - Prediction interval based on t-distribution (df = 15)
## - Logit transformation
```

SUBGROUP ANALYSIS UK vs Non UK

```r
table(df_primary_logit$Country, useNA = "ifany")
```

```
##
##      Brazil      Canada       China      France       Italy Netherlands
##           1           1           1           1           2           1
##       Spain          UK         USA
##           1          10           1
```

```r
library(dplyr)
library(meta)

# Collapse to UK vs non-UK
df_primary_logit <- df_primary_logit %>%
  mutate(UK_group = ifelse(Country == "UK", "UK", "Non-UK"))

table(df_primary_logit$UK_group)
```

```
##
## Non-UK     UK
##      9     10
```

```r
primary_subgroup_uk <- metagen(
  TE               = logit_c,
  seTE             = se_logit,
  studlab          = Index,
  data             = df_primary_logit,
  sm               = "PLOGIT",
  method.tau       = "REML",
  method.random.ci = "HK",
  subgroup         = UK_group,
  tau.common       = TRUE,
  prediction       = TRUE
)
summary(primary_subgroup_uk)
```

```
##     proportion           95% CI %W(common) %W(random) UK_group
## 1       0.6630 [0.6453; 0.6803]        2.8        6.3   Non-UK
## 3       0.7600 [0.5001; 0.9093]        0.0        1.5   Non-UK
## 6a      0.7400 [0.7195; 0.7595]        1.6        6.2       UK
## 6b      0.7000 [0.6361; 0.7569]        0.2        5.3       UK
## 6c      0.7160 [0.6904; 0.7403]        1.1        6.1       UK
```

```
## 7       0.7030 [0.5902; 0.7955]        0.1        3.9   Non-UK
## 8a      0.7520 [0.7296; 0.7731]        1.3        6.2       UK
## 8b      0.7940 [0.7694; 0.8166]        0.8        6.1       UK
## 8c      0.7640 [0.7379; 0.7883]        0.9        6.1       UK
## 8d      0.8150 [0.7911; 0.8367]        0.8        6.0       UK
## 16      0.8780 [0.8346; 0.9112]        0.1        4.8   Non-UK
## 19b     0.6640 [0.6600; 0.6680]       53.7        6.4       UK
## 23      0.7200 [0.6995; 0.7396]        1.7        6.2   Non-UK
## 26b     0.6700 [0.6287; 0.7088]        0.5        5.9   Non-UK
## 29b     0.6600 [0.4487; 0.8224]        0.0        2.2   Non-UK
## 35      0.8660 [0.7660; 0.9273]        0.0        2.9   Non-UK
## 36      0.6740 [0.6690; 0.6790]       33.3        6.4       UK
## 39      0.7440 [0.7128; 0.7729]        0.7        6.0       UK
## 40      0.7900 [0.7468; 0.8276]        0.3        5.5   Non-UK
##
## Number of studies: k = 19
##
##                          proportion           95% CI    z|t  p-value
## Common effect model          0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)    0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                 [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1107 [0.0544; 0.2972]; tau = 0.3327 [0.2333; 0.5452]
##  I^2 = 96.0% [97.5%; 99.5%]; H = 4.99 [6.37; 14.75]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Results for subgroups (common effect model):
##                     k proportion           95% CI      Q   I^2
## UK_group = Non-UK   9     0.6981 [0.6864; 0.7097]  86.45 90.7%
## UK_group = UK      10     0.6753 [0.6723; 0.6782] 337.07 97.3%
##
## Test for subgroup differences (common effect model):
##                     Q d.f.  p-value
## Between groups  13.44    1   0.0002
## Within groups  423.52   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                     k proportion           95% CI  tau^2    tau
## UK_group = Non-UK   9     0.7505 [0.6728; 0.8149] 0.1107 0.3327
## UK_group = UK      10     0.7388 [0.7017; 0.7728] 0.1107 0.3327
##
## Test for subgroup differences (random effects model (HK)):
##                     Q d.f.  p-value
## Between groups   0.11    1   0.7372
## Within groups  423.52   17 < 0.0001
##
```

```
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

SUGGROUP ANALYSIS - FOLLOW UP

```r
# See the follow-up values across the pool
df_primary_logit %>%
  select(Index, Disease_Group, `Follow-Up`, Estimate) %>%
  arrange(`Follow-Up`) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 4
##    Index Disease_Group                      `Follow-Up`                 Estimate
##    <chr> <chr>                              <chr>                          <dbl>
##  1 6a    Psoriasis                          "10 years"                     0.74
##  2 6b    Psoriatic Arthritis (PsA)          "10 years"                     0.7
##  3 6c    Rheumatoid Arthritis (RA)          "10 years"                     0.716
##  4 7     Systemic Lupus Erythematosus (SLE) "10 years"                     0.703
##  5 8a    Rheumatoid Arthritis (RA)          "10 years"                     0.752
##  6 8b    Ankylosing Spondylitis (AS)        "10 years"                     0.794
##  7 8c    Psoriatic Arthritis (PsA)          "10 years"                     0.764
##  8 8d    Psoriasis                          "10 years"                     0.815
##  9 19b   Diabetes (T2DM)                    "10 years"                     0.664
## 10 26b   Systemic Lupus Erythematosus (SLE) "10 years"                     0.67
## 11 29b   Ankylosing Spondylitis (AS)        "10 years"                     0.66
## 12 35    Psoriatic Arthritis (PsA)          "10 years"                     0.866
## 13 1     Diabetes (T2DM)                    "5 years"                      0.663
## 14 36    Diabetes (T2DM)                    "5 years (4.9 years)"          0.674
## 15 39    Rheumatoid Arthritis (RA)          "6.0 years"                    0.744
## 16 3     Systemic Lupus Erythematosus (SLE) "7 years (IQR 3-9 years)"      0.76
## 17 16    Diabetes (T2DM)                    "8.68 ± 1.86years."            0.878
## 18 40    Rheumatoid Arthritis (RA)          "9,957 patient-years of fo~    0.79
## 19 23    Diabetes (T2DM)                    "Up to 5 years"                0.72
```

```r
library(dplyr)
library(meta)

fu_under10 <- c("1", "36", "3", "16", "40", "23")

df_primary_logit <- df_primary_logit %>%
  mutate(Follow_group = ifelse(Index %in% fu_under10, "Under 10y", "10y or more"))

table(df_primary_logit$FU_group)   #
```

```
## Warning: Unknown or uninitialised column: `FU_group`.

## < table of extent 0 >
```

```r
sens_followup <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = df_primary_logit,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  subgroup = Follow_group,
  tau.common = TRUE,
  prediction = TRUE
)
summary(sens_followup)
```

```
##     proportion           95% CI %W(common) %W(random) Follow_group
## 1       0.6630 [0.6453; 0.6803]        2.8        6.3    Under 10y
## 3       0.7600 [0.5001; 0.9093]        0.0        1.5    Under 10y
## 6a      0.7400 [0.7195; 0.7595]        1.6        6.2  10y or more
## 6b      0.7000 [0.6361; 0.7569]        0.2        5.3  10y or more
## 6c      0.7160 [0.6904; 0.7403]        1.1        6.1  10y or more
## 7       0.7030 [0.5902; 0.7955]        0.1        3.9  10y or more
## 8a      0.7520 [0.7296; 0.7731]        1.3        6.2  10y or more
## 8b      0.7940 [0.7694; 0.8166]        0.8        6.1  10y or more
## 8c      0.7640 [0.7379; 0.7883]        0.9        6.1  10y or more
## 8d      0.8150 [0.7911; 0.8367]        0.8        6.0  10y or more
## 16      0.8780 [0.8346; 0.9112]        0.1        4.8    Under 10y
## 19b     0.6640 [0.6600; 0.6680]       53.7        6.4  10y or more
## 23      0.7200 [0.6995; 0.7396]        1.7        6.2    Under 10y
## 26b     0.6700 [0.6287; 0.7088]        0.5        5.9  10y or more
## 29b     0.6600 [0.4487; 0.8224]        0.0        2.2  10y or more
## 35      0.8660 [0.7660; 0.9273]        0.0        2.9  10y or more
## 36      0.6740 [0.6690; 0.6790]       33.3        6.4    Under 10y
## 39      0.7440 [0.7128; 0.7729]        0.7        6.0  10y or more
## 40      0.7900 [0.7468; 0.8276]        0.3        5.5    Under 10y
##
## Number of studies: k = 19
##
##                             proportion           95% CI    z|t  p-value
## Common effect model             0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)       0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                    [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1113 [0.0546; 0.2995]; tau = 0.3337 [0.2337; 0.5473]
##  I^2 = 96.1% [94.5%; 98.9%]; H = 5.07 [4.25; 9.74]
```

```
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
##
## Results for subgroups (common effect model):
##                              k proportion           95% CI      Q   I^2
## Follow_group = Under 10y     6     0.6774 [0.6727; 0.6820]  89.11 94.4%
## Follow_group = 10y or more  13     0.6761 [0.6724; 0.6797] 347.66 96.5%
##
## Test for subgroup differences (common effect model):
##                     Q d.f.  p-value
## Between groups   0.19    1   0.6611
## Within groups  436.76   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                              k proportion           95% CI  tau^2    tau
## Follow_group = Under 10y     6     0.7480 [0.6396; 0.8324] 0.1113 0.3337
## Follow_group = 10y or more  13     0.7415 [0.7040; 0.7758] 0.1113 0.3337
##
## Test for subgroup differences (random effects model (HK)):
##                     Q d.f.  p-value
## Between groups   0.02    1   0.8746
## Within groups  436.76   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

DESCRIPTIVE ANALYSIS - HELLERS K

```r
library(dplyr)

carma <- df_all %>%
  filter(Index %in% c("5a", "5b", "5c")) %>%
  select(Index, Disease_Group, Country, Sample_size,
         Estimate, LowerCI, UpperCI, Dscrimination, `Follow-Up`)

carma
```

```
## # A tibble: 3 x 9
##   Index Disease_Group Country Sample_size Estimate LowerCI UpperCI Dscrimination
##   <chr> <chr>         <chr>         <dbl>    <dbl>   <dbl>   <dbl> <chr>
## 1 5a    "\r\n Rheuma~ Spain           708    0.607      NA      NA Heller's K
## 2 5b    "Ankylosing ~ Spain           692    0.592      NA      NA Heller's K
## 3 5c    "Psoriatic A~ Spain           680    0.58       NA      NA Heller's K
```

```
## # i 1 more variable: `Follow-Up` <chr>
```

SENSITIVITY ANALYSIS - CROSS-SECTIONAL

```r
library(dplyr)

# Pulling the candidate cross-sectional surrogate-outcome studies from df_all
cross_sec <- df_all %>%
  filter(Outcome_type != "Appropriate") %>%
  select(Index, Disease_Group, Outcome_type, Outcome_Description,
         `Follow-Up`, Estimate, LowerCI, UpperCI, Dscrimination)

cross_sec
```

```
## # A tibble: 9 x 9
##   Index Disease_Group      Outcome_type Outcome_Description `Follow-Up` Estimate
##   <chr> <chr>              <chr>        <chr>               <chr>          <dbl>
## 1 4     Diabetes (T1DM & ~ Somewhat_Ap~ "CVD( not includin~ "1 - 12 ye~    0.788
## 2 14    Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "3.8 month~    0.754
## 3 18    Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~    0.765
## 4 21    Metabolic Syndrom~ Not_Appropr~ "elevated cfPWV"    "Cross-sec~    0.697
## 5 22    Inflammatory Bowe~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~    0.812
## 6 27b   Systemic Lupus Er~ Not_Appropr~ "carotid plaque fo~ "Cross-sec~    0.82
## 7 28    Rheumatoid Arthri~ Not_Appropr~ "carotid plaque fo~ "cross-sec~    0.64
## 8 32    chronic kidney di~ Not_Appropr~ "HbA1c and carotid~ "cross-sec~    0.51
## 9 37    Psoriatic Arthrit~ Not_Appropr~ "carotid plaque fo~ "cross-sec~    0.69
## # i 3 more variables: LowerCI <dbl>, UpperCI <dbl>, Dscrimination <chr>
```

```r
library(dplyr)

cross_sec <- df_all %>%
  filter(Index %in% c("14","18","21","22","28","32","37","27b")) %>%
  select(Index, Disease_Group, Outcome_Description, Dscrimination,
         Estimate, LowerCI, UpperCI, `Follow-Up`, Confidence_interval)

cross_sec
```

```
## # A tibble: 8 x 9
##   Index Disease_Group Outcome_Description Dscrimination Estimate LowerCI UpperCI
##   <chr> <chr>         <chr>               <chr>            <dbl>   <dbl>   <dbl>
## 1 14    Systemic Lup~ carotid plaque for~ AUC              0.754   0.64    0.87
## 2 18    Systemic Lup~ carotid plaque for~ AUC              0.765   0.711   0.82
## 3 21    Metabolic Sy~ elevated cfPWV      AUC              0.697   0.659   0.736
## 4 22    Inflammatory~ carotid plaque for~ AUC              0.812   0.748   0.875
## 5 27b   Systemic Lup~ carotid plaque for~ AUC              0.82    0.76    0.89
## 6 28    Rheumatoid A~ carotid plaque for~ AUC              0.64    0.53    0.75
## 7 32    chronic kidn~ HbA1c and carotid ~ AUC              0.51    0.32    0.7
## 8 37    Psoriatic Ar~ carotid plaque for~ AUC              0.69    0.6     0.77
## # i 2 more variables: `Follow-Up` <chr>, Confidence_interval <chr>
```

```r
# Discrimination for subclinical carotid atherosclerosis
library(dplyr)
library(meta)

plaque <- df_all %>%
  filter(Index %in% c("14","18","22","28","37","27b")) %>%   # carotid plaque only; 21 &
  ↪  32 excluded
  mutate(
    logit_c  = log(Estimate / (1 - Estimate)),
    logit_lb = log(LowerCI  / (1 - LowerCI)),
    logit_ub = log(UpperCI  / (1 - UpperCI)),
    se_logit = (logit_ub - logit_lb) / (2 * 1.96)
  )

nrow(plaque)
```

```
## [1] 6
```

```r
pool_plaque <- metagen(
  TE = logit_c,
  seTE = se_logit,
  studlab = Index,
  data = plaque,
  sm = "PLOGIT",
  method.tau = "REML",
  method.random.ci = "HK",
  prediction = TRUE
)
summary(pool_plaque)
```

```
##     proportion           95% CI %W(common) %W(random)
## 14      0.7540 [0.6124; 0.8560]        6.9       11.4
## 18      0.7650 [0.7052; 0.8158]       31.8       21.2
## 22      0.8120 [0.7377; 0.8690]       16.4       17.4
## 27b     0.8200 [0.7403; 0.8793]       13.7       16.2
## 28      0.6400 [0.5215; 0.7436]       12.6       15.6
## 37      0.6900 [0.5984; 0.7688]       18.7       18.2
##
## Number of studies: k = 6
##
##                              proportion           95% CI    z|t  p-value
## Common effect model              0.7542 [0.7206; 0.7850] 12.66 < 0.0001
## Random effects model (HK)        0.7527 [0.6747; 0.8171]  7.45   0.0007
## Prediction interval                     [0.5719; 0.8740]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.0804 [0.0000; 0.7589]; tau = 0.2835 [0.0000; 0.8711]
##  I^2 = 60.2% [2.2%; 83.8%]; H = 1.58 [1.01; 2.48]
##
## Test of heterogeneity:
##      Q d.f. p-value
##  12.55    5  0.0280
```

```
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 5)
## - Prediction interval based on t-distribution (df = 5)
## - Logit transformation
```

```
df_primary_logit %>%
  select(Index, Sample_size, Study_Year, `Follow-Up`
        ) %>%
  print(n = Inf)
```

```
## # A tibble: 19 x 4
##    Index Sample_size Study_Year `Follow-Up`
##    <chr>       <dbl>      <dbl> <chr>
##  1 1           20793       2025 "5 years"
##  2 3             143       2025 "7 years (IQR 3-9 years)"
##  3 6a           8062       2025 "10 years"
##  4 6b            769       2025 "10 years"
##  5 6c           4772       2025 "10 years"
##  6 7             550       2025 "10 years"
##  7 8a          91750       2023 "10 years"
##  8 8b          42306       2023 "10 years"
##  9 8c          26375       2023 "10 years"
## 10 8d         390664       2023 "10 years"
## 11 16            566       2022 "8.68 ± 1.86years."
## 12 19b        168871       2022 "10 years"
## 13 23           6245       2021 "Up to 5 years"
## 14 26b          1887       2021 "10 years"
## 15 29b           133       2020 "10 years"
## 16 35            155       2018 "10 years"
## 17 36         181399       2018 "5 years (4.9 years)"
## 18 39          12747       2017 "6.0 years"
## 19 40           1050       2013 "9,957 patient-years of follow-up\r\n9.5 years"
```

SUBGROUP ANALYSIS - MULTI-VS-SINGLE

```
library(meta)

# First check how the groups split and that the labels are clean
table(df_primary_logit$`Single_vs_multi-centre`, useNA = "ifany")
```

```
##
##  Multi-centre Single-centre
##            14             5
```

```r
primary_subgroup_centre <- metagen(
  TE                = logit_c,
  seTE              = se_logit,
  studlab           = Index,
  data              = df_primary_logit,
  sm                = "PLOGIT",
  method.tau        = "REML",
  method.random.ci = "HK",
  subgroup          = `Single_vs_multi-centre`,
  tau.common        = TRUE,
  prediction        = TRUE
)
summary(primary_subgroup_centre)
```

```
##     proportion           95% CI %W(common) %W(random) Single_vs_multi-centre
## 1       0.6630 [0.6453; 0.6803]        2.8        6.3           Multi-centre
## 3       0.7600 [0.5001; 0.9093]        0.0        1.5          Single-centre
## 6a      0.7400 [0.7195; 0.7595]        1.6        6.2           Multi-centre
## 6b      0.7000 [0.6361; 0.7569]        0.2        5.3           Multi-centre
## 6c      0.7160 [0.6904; 0.7403]        1.1        6.1           Multi-centre
## 7       0.7030 [0.5902; 0.7955]        0.1        3.9          Single-centre
## 8a      0.7520 [0.7296; 0.7731]        1.3        6.2           Multi-centre
## 8b      0.7940 [0.7694; 0.8166]        0.8        6.1           Multi-centre
## 8c      0.7640 [0.7379; 0.7883]        0.9        6.1           Multi-centre
## 8d      0.8150 [0.7911; 0.8367]        0.8        6.0           Multi-centre
## 16      0.8780 [0.8346; 0.9112]        0.1        4.8          Single-centre
## 19b     0.6640 [0.6600; 0.6680]       53.7        6.4           Multi-centre
## 23      0.7200 [0.6995; 0.7396]        1.7        6.2          Single-centre
## 26b     0.6700 [0.6287; 0.7088]        0.5        5.9          Single-centre
## 29b     0.6600 [0.4487; 0.8224]        0.0        2.2           Multi-centre
## 35      0.8660 [0.7660; 0.9273]        0.0        2.9           Multi-centre
## 36      0.6740 [0.6690; 0.6790]       33.3        6.4           Multi-centre
## 39      0.7440 [0.7128; 0.7729]        0.7        6.0           Multi-centre
## 40      0.7900 [0.7468; 0.8276]        0.3        5.5           Multi-centre
##
## Number of studies: k = 19
##
##                              proportion           95% CI    z|t  p-value
## Common effect model              0.6766 [0.6737; 0.6794] 110.13        0
## Random effects model (HK)        0.7433 [0.7089; 0.7749]  12.89 < 0.0001
## Prediction interval                     [0.5912; 0.8529]
##
## Quantifying heterogeneity (with 95% CIs):
##  tau^2 = 0.1026 [0.0515; 0.2737]; tau = 0.3203 [0.2269; 0.5232]
##  I^2 = 95.9% [94.6%; 96.8%]; H = 4.93 [4.32; 5.62]
##
## Quantifying residual heterogeneity (with 95% CIs):
##  tau^2 = 0.1100 [0.0544; 0.2997]; tau = 0.3317 [0.2333; 0.5475]
##  I^2 = 95.9% [97.6%; 99.6%]; H = 4.92 [6.48; 15.07]
##
## Test of heterogeneity:
##       Q d.f.  p-value
##  436.96   18 < 0.0001
```

```
##
## Results for subgroups (common effect model):
##                                          k proportion           95% CI      Q
## Single_vs_multi-centre = Multi-centre   14     0.6754 [0.6725; 0.6783] 372.44
## Single_vs_multi-centre = Single-centre   5     0.7211 [0.7041; 0.7376]  38.95
##                                          I^2
## Single_vs_multi-centre = Multi-centre  96.5%
## Single_vs_multi-centre = Single-centre 89.7%
##
## Test for subgroup differences (common effect model):
##                      Q d.f.  p-value
## Between groups   25.57    1 < 0.0001
## Within groups   411.39   17 < 0.0001
##
## Results for subgroups (random effects model (HK)):
##                                          k proportion           95% CI  tau^2
## Single_vs_multi-centre = Multi-centre   14     0.7412 [0.7045; 0.7748] 0.1100
## Single_vs_multi-centre = Single-centre   5     0.7513 [0.6107; 0.8533] 0.1100
##                                           tau
## Single_vs_multi-centre = Multi-centre  0.3317
## Single_vs_multi-centre = Single-centre 0.3317
##
## Test for subgroup differences (random effects model (HK)):
##                      Q d.f.  p-value
## Between groups    0.05    1   0.8315
## Within groups   411.39   17 < 0.0001
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
##   (assuming common tau^2 in subgroups)
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 18)
## - Prediction interval based on t-distribution (df = 18)
## - Logit transformation
```

```r
data.frame(
  Q  = primary_subgroup_centre$Q.b.random,
  df = primary_subgroup_centre$df.Q.b,
  p  = primary_subgroup_centre$pval.Q.b.random
)
```

```
##           Q df         p
## 1 0.0452603  1 0.8315261
```

```r
data.frame(
  centre = primary_subgroup_centre$subgroup.levels,
  k      = primary_subgroup_centre$k.w,
  c_stat = plogis(primary_subgroup_centre$TE.random.w),
  lower  = plogis(primary_subgroup_centre$lower.random.w),
  upper  = plogis(primary_subgroup_centre$upper.random.w),
  I2     = primary_subgroup_centre$I2.w
```

```r
)
```

```
##                        centre  k    c_stat     lower     upper        I2
## Multi-centre   Multi-centre 14 0.7411854 0.7044886 0.7747820 0.9650948
## Single-centre Single-centre  5 0.7512874 0.6107153 0.8532924 0.8973115
```

METAREGRESSION

Event meta regression

```r
library(metafor)

# Log-transform events
df_primary_logit$log_events <- log(df_primary_logit$Events)

# Refit the primary model in metafor as the reference
m_base <- rma(
  yi = logit_c,
  sei = se_logit,
  data = df_primary_logit,
  method = "REML")
m_base
```

```
##
## Random-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of total heterogeneity): 0.1026 (SE = 0.0401)
## tau (square root of estimated tau^2 value):      0.3203
## I^2 (total heterogeneity / total variability):   98.70%
## H^2 (total variability / sampling variability):  77.03
##
## Test for Heterogeneity:
## Q(df = 18) = 436.9564, p-val < .0001
##
## Model Results:
##
## estimate      se     zval    pval   ci.lb   ci.ub
##   1.0631  0.0809  13.1369  <.0001  0.9045  1.2217  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Univariable meta-regression on log(events)
mr_events <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ log_events,
  data = df_primary_logit,
  method = "REML")
```

```r
mr_events
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.1073 (SE = 0.0430)
## tau (square root of estimated tau^2 value):             0.3275
## I^2 (residual heterogeneity / unaccounted variability): 98.58%
## H^2 (unaccounted variability / sampling variability):   70.35
## R^2 (amount of heterogeneity accounted for):            0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 294.7968, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3705, p-val = 0.5427
##
## Model Results:
##
##             estimate      se     zval    pval    ci.lb   ci.ub
## intrcpt       1.2189  0.2679   4.5496  <.0001   0.6938  1.7440  ***
## log_events   -0.0230  0.0378  -0.6087  0.5427  -0.0971  0.0511
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


```r
mr_events
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.1073 (SE = 0.0430)
## tau (square root of estimated tau^2 value):             0.3275
## I^2 (residual heterogeneity / unaccounted variability): 98.58%
## H^2 (unaccounted variability / sampling variability):   70.35
## R^2 (amount of heterogeneity accounted for):            0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 294.7968, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.3705, p-val = 0.5427
##
## Model Results:
##
##             estimate      se     zval    pval    ci.lb   ci.ub
## intrcpt       1.2189  0.2679   4.5496  <.0001   0.6938  1.7440  ***
## log_events   -0.0230  0.0378  -0.6087  0.5427  -0.0971  0.0511
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

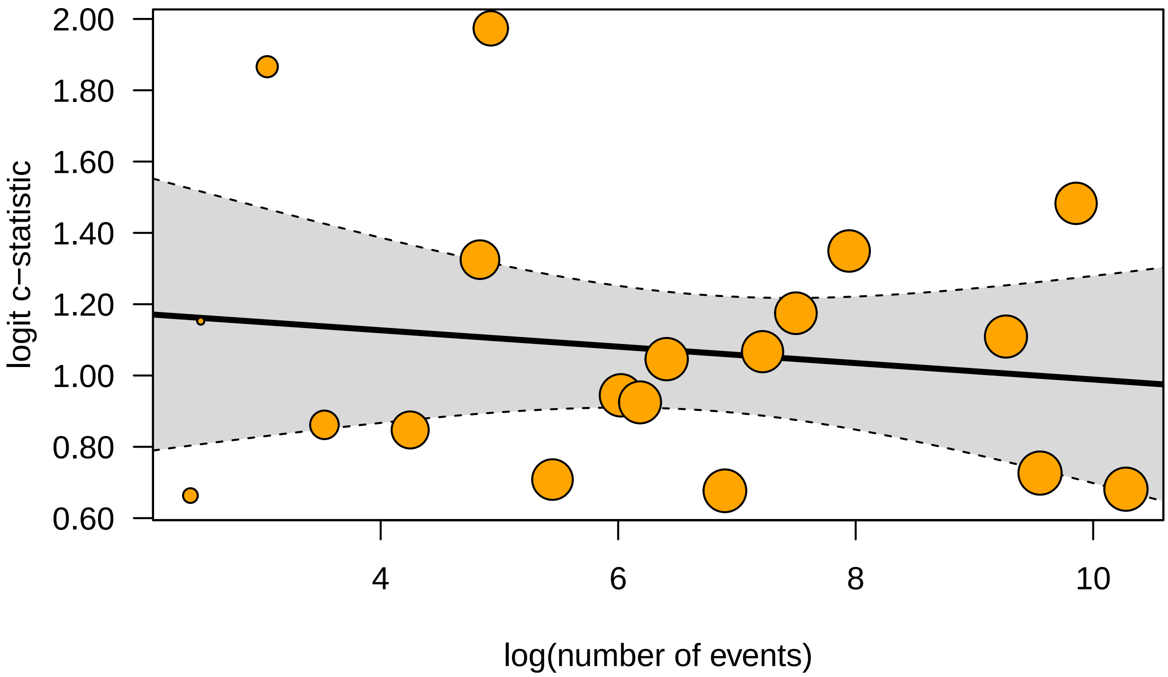
To make the slope interpretation on the c-statistic scale

```
# Predicted c-statistic across the range of events
# use the fitted metaregression model to estimate the c-statistic for studies at the
↪  event number
newpts <- data.frame(log_events = log(c(20, 100, 500, 5000, 25000))) # create dataframe
↪  for hypothetical moderators/event number
prediction <- predict( # function to calculate predicted values from fitted model
  mr_events, # fitted regression model
  newmods = as.matrix(newpts$log_events), # convert moderator to a matrix to feed to
   ↪  function
  transf = plogis
  )

# build result table
data.frame(events = c(20, 100, 500, 5000, 25000),
           c_stat = round(prediction$pred, 3),
           lower  = round(prediction$ci.lb, 3),
           upper  = round(prediction$ci.ub, 3))
```

```
##   events c_stat lower upper
## 1     20  0.760 0.696 0.813
## 2    100  0.753 0.708 0.792
## 3    500  0.746 0.713 0.776
## 4   5000  0.736 0.693 0.774
## 5  25000  0.728 0.665 0.783
```

Bubble plot for the figure

```
regplot(mr_events,
        mod = "log_events", # name of relevant data in the model object
        xlab = "log(number of events)",
        ylab = "logit c-statistic",
        las = 1, # axis label
        digits = 2, # number of decimal places in the plot
        bg = "orange")
```



Event rate - meta regression

```
df_primary_logit$event_rate <- df_primary_logit$Events / df_primary_logit$Sample_size

mr_rate <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ event_rate,
  data = df_primary_logit,
  method = "REML")

mr_rate
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.1016 (SE = 0.0410)
## tau (square root of estimated tau^2 value):             0.3188
## I^2 (residual heterogeneity / unaccounted variability): 96.93%
## H^2 (unaccounted variability / sampling variability):   32.53
## R^2 (amount of heterogeneity accounted for):            0.94%
##
## Test for Residual Heterogeneity:
```

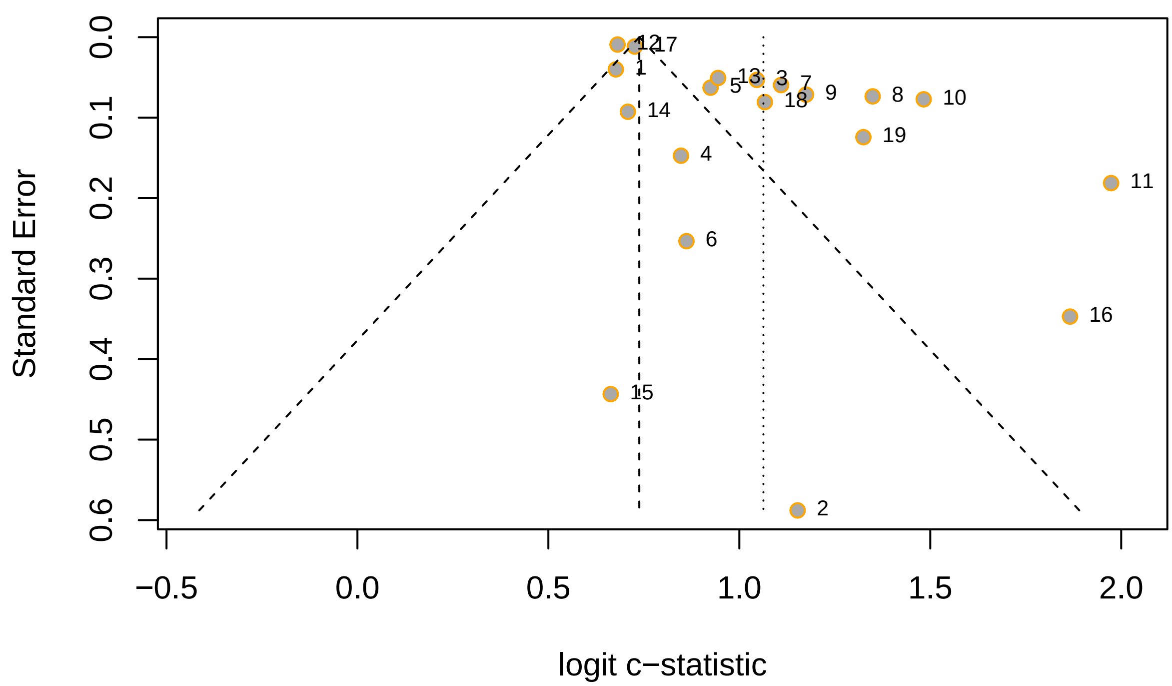
```
## QE(df = 17) = 378.5674, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 2.3908, p-val = 0.1221
##
## Model Results:
##
##             estimate      se    zval    pval    ci.lb   ci.ub
## intrcpt       0.7954  0.1909  4.1677  <.0001   0.4213  1.1695  ***
## event_rate    2.7159  1.7565  1.5462  0.1221  -0.7267  6.1585
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Sample Size - meta regression

```
df_primary_logit$log_n <- log(df_primary_logit$Sample_size)

mr_sample_size <- rma(
  yi = logit_c,
  sei = se_logit,
  mods = ~ log_n,
  data = df_primary_logit,
  method = "REML")

mr_sample_size
```

```
##
## Mixed-Effects Model (k = 19; tau^2 estimator: REML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.1061 (SE = 0.0425)
## tau (square root of estimated tau^2 value):             0.3258
## I^2 (residual heterogeneity / unaccounted variability): 98.70%
## H^2 (unaccounted variability / sampling variability):   77.06
## R^2 (amount of heterogeneity accounted for):            0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 17) = 318.3170, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.6228, p-val = 0.4300
##
## Model Results:
##
##          estimate      se     zval    pval    ci.lb   ci.ub
## intrcpt    1.3251  0.3414   3.8813  0.0001   0.6560  1.9943  ***
## log_n     -0.0286  0.0362  -0.7892  0.4300  -0.0995  0.0424
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(meta)

# Build the numbered key from the pooled object (row order = plot order)
key <- data.frame(
  No     = seq_along(primary_logit$studlab), # create number sequence
  Study = primary_logit$studlab,
  c_stat = round(plogis(primary_logit$TE), 3) # study effect column
)

# Funnel plot labelled with numbers
meta::funnel(primary_logit,
             studlab      = as.character(key$No),# display numbers
             cex.studlab = 0.7, # font size
             pos.studlab = 4, # label to the right of each point
             xlab         = "logit c-statistic",
             backtransf   = FALSE,
             col = "orange"
)
```




```r
# Print the mapping to place in the figure caption or legend
print(key, row.names = FALSE)
```

```
##  No                            Study c_stat
##   1 Enguita-German et al. (T2DM-2025)  0.663
##   2         Azoulay et al. (SLE-2025)  0.760
```

```
##   3           Hughes et al. (Ps-2025)  0.740
##   4          Hughes et al. (PsA-2025)  0.700
##   5           Hughes et al. (RA-2025)  0.716
##   6         Rosário et al. (SLE-2025)  0.703
##   7           Hughes et al. (RA-2023)  0.752
##   8           Hughes et al. (AS-2023)  0.794
##   9          Hughes et al. (PsA-2023)  0.764
##  10           Hughes et al. (Ps-2023)  0.815
##  11             Mu et al. (T2DM-2022)  0.878
##  12    Dziopa K et al. (T2DM-Q3-2022)  0.664
##  13           Hong et al. (T2DM-2021)  0.720
##  14  Sivakumaran et al. (SLE-Q3-2021)  0.670
##  15      Navarini et al. (AS-Q3-2020)  0.660
##  16        Navarini et al. (PsA-2018)  0.866
##  17           Read et al. (T2DM-2018)  0.674
##  18            Alemao et al.(RA-2017)  0.744
##  19         Arts EEA et al. (RA-2013)  0.790
```

```
# Egger-type test for funnel asymmetry
metabias(primary_logit, method.bias = "linreg", k.min = 10)
```

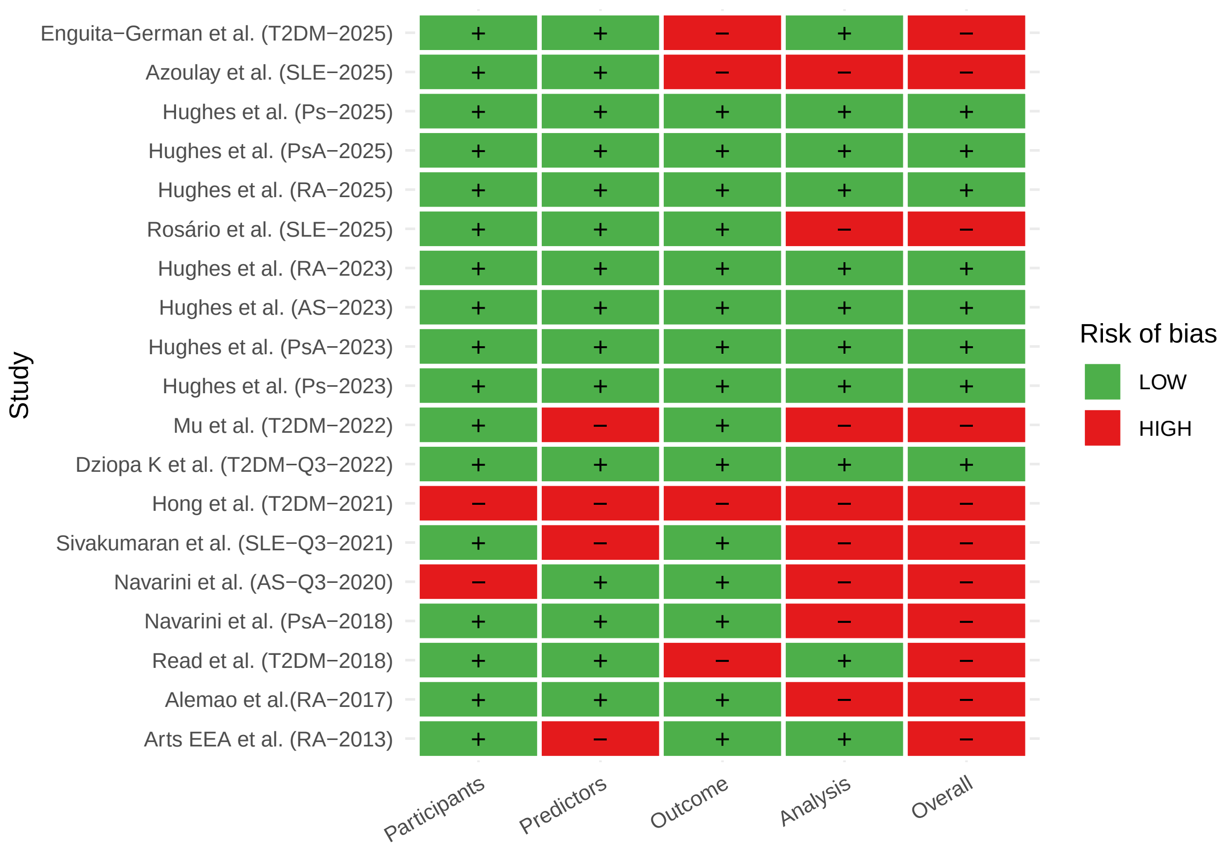
```
## Linear regression test of funnel plot asymmetry
##
## Test result: t = 4.63, df = 17, p-value = 0.0002
## Bias estimate: 4.4525 (SE = 0.9622)
##
## Details:
## - multiplicative residual heterogeneity variance (tau^2 = 11.3756)
## - predictor: standard error
## - weight:    inverse variance
## - reference: Egger et al. (1997), BMJ
```

```
paste(paste0(key$No, " = ", key$Study), collapse = "; ") # caption for keys
```

```
## [1] "1 = Enguita-German et al. (T2DM-2025); 2 = Azoulay et al. (SLE-2025); 3 = Hughes et al. (Ps-2025
```

```
#traffic light plot
library(dplyr); library(tidyr); library(ggplot2)

rob <- df_primary_logit %>%
  select(Index, Author, Participants, Predictors2, Outcome3, Analysis, Overall) %>%
  pivot_longer(c(-Index, -Author), names_to = "Domain", values_to = "Rating") %>%
  mutate(
    Domain = recode(Domain, Predictors2 = "Predictors", Outcome3 = "Outcome"),
    Domain = factor(Domain,
             levels = c("Participants","Predictors","Outcome","Analysis","Overall")),
    Rating = factor(toupper(trimws(Rating)), levels = c("LOW","UNCLEAR","HIGH")))

ggplot(rob, aes(x = Domain, y = factor(Author, levels = rev(unique(Author))))) +
  geom_tile(aes(fill = Rating), colour = "white", linewidth = 0.8) +
  geom_text(aes(label = recode(as.character(Rating),
              LOW = "+", UNCLEAR = "?", HIGH = "\u2212")), size = 3.5) +
```

```
  scale_fill_manual(values = c(LOW = "#4DAF4A", UNCLEAR = "#FFD92F", HIGH = "#E41A1C")) +
  labs(x = NULL, y = "Study", fill = "Risk of bias") +
  theme_minimal(base_size = 10) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```




```
png("metareg_panel.png", width = 2400, height = 900, res = 200)
par(mfrow = c(1, 3), mar = c(5, 5, 3, 2))
regplot(mr_events, mod = "log_events", xlab = "log(events)",
        ylab = "logit c-statistic", main = "A", bg = "grey")
regplot(mr_sample_size, mod = "log_n", xlab = "log(sample size)",
        ylab = "", main = "B", bg = "grey")
regplot(mr_rate, mod = "event_rate", xlab = "Event rate",
        ylab = "", main = "C", bg = "grey")
dev.off()
```

```
## pdf
##   2
```